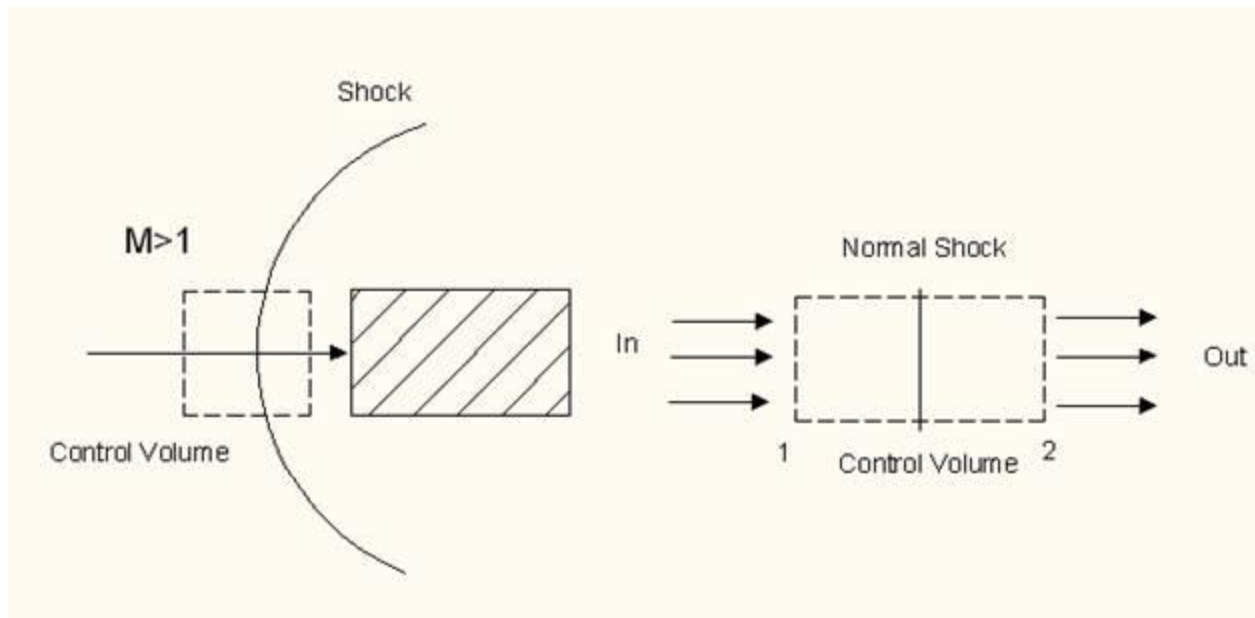


Normal shock relations

It had already been discussed that the subsonic flow is pre-warned and supersonic flow is not. The reason behind this fact is that, any small amplitude disturbance travels with acoustic speed, however speed of fluid particle is more than the speed of sound in case of supersonic flows. Therefore the message of presence of the obstacle cannot propagate upstream. Hence a messenger gets developed in front of the obstacle to warn the flow in order to avoid its direct collision with the obstacle. This messenger is called as shock. In the presence of normal shock, fluid velocity decreases to the extent where flow Mach number behind the shock attains value below one. Due to this subsonic speed attainment of the flow, it becomes aware about the presence of the obstacle well in advance in the narrow space between shock and obstacle. Herewith we will deal for computation of flow properties behind the normal shock.

In the presence of a general obstacle the shock pattern is shown here in **Fig.**



Shock pattern for a blunt or bluff obstacle

The shock for the stagnation streamline can be considered as normal to it. Therefore we can use the earlier derived 1D flow relations along with the assumptions of flow steady, adiabatic and inviscid flow. Consider a small control volume around normal shock for application of these relations between two stations of the control volume, mainly, inlet and outlet as shown in **Fig.**

$$\rho_1 u_1 = \rho_2 u_2$$

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$$

Lets us the reference star properties of the flow in the process to calculate the flow properties behind the normal shock from the known inlet conditions. We can take the advantage of using stared temperature since the flow is adiabatic in nature. Imagine that flow is adiabatically brought to Mach number one on either sides of the shock independantly. In this case, we should get same stared temperature on either sides of shock. We can also show that total temperature is also same on either sides. The explicit formulation using the star temperature and concerned acoustic speed before the normal shock is,

$$h_1 + \frac{u_1^2}{2} = h_1^* + \frac{u_1^*}{2}$$

$$h_1 + \frac{u_1^2}{2} = C_p T^* + \frac{a^*}{2}$$

$$h_1 + \frac{u_1^2}{2} = C_p T^* + \frac{\gamma R T^*}{2} = \frac{\gamma R T^*}{\gamma - 1} + \frac{\gamma R T^*}{2} \left[Q C_p = \frac{\gamma R}{\gamma - 1} \right]$$

$$h_1 + \frac{u_1^2}{2} = \gamma \left[\frac{1}{\gamma - 1} + \frac{1}{2} \right] R T^* = \frac{\gamma(\gamma + 1)}{2(\gamma - 1)} R T^* = \frac{(\gamma + 1)}{2(\gamma - 1)} a^{*2}$$

Applying same strategy at the outlet we get,

$$h_2 + \frac{u_2^2}{2} = \frac{(\gamma + 1)}{2(\gamma - 1)} a^{*2}$$

However, we can write static enthalpy in terms of acoustic speed as,

$$h = C_p T = \frac{\gamma R}{\gamma - 1} T = \frac{\gamma R T}{\gamma - 1} = \frac{a^2}{\gamma - 1}$$

Therefore, the energy equation at the inlet becomes,

$$a_1^2 = \frac{(\gamma + 1)}{2} a^{*2} - \frac{(\gamma - 1)}{2} u_1^2$$

Similarly for the outlet station we have

$$a_2^2 = \frac{(\gamma + 1)}{2} a^{*2} - \frac{(\gamma - 1)}{2} u_2^2$$

Let's obtain the expression for velocity using mass and momentum equations to replace the acoustic speed term from equations (6.4) and (6.5).

From 1D mass and momentum conservation equations we have

$$\rho_1 u_1 = \rho_2 u_2$$

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

Therefore,

$$\frac{p_1 + \rho_1 u_1^2}{\rho_1 u_1} = \frac{p_2 + \rho_2 u_2^2}{\rho_2 u_2}$$

$$\frac{p_1}{\rho_1 u_1} + u_1 = \frac{p_2}{\rho_2 u_2} + u_2 \quad [Q \rho_1 u_1 = \rho_2 u_2 \text{ mass conservation}]$$

$$\frac{a_1^2}{\gamma u_1} + u_1 = \frac{a_2^2}{\gamma u_2} + u_2$$

$$u_2 - u_1 = \frac{a_1^2}{\gamma u_1} - \frac{a_2^2}{\gamma u_2}$$

Using equation 6.4 and 6.5, above equation transforms to

$$u_2 - u_1 = \frac{1}{\gamma u_1} \left[\frac{(\gamma+1)}{2} a^{*2} - \frac{(\gamma-1)}{2} u_1^2 \right] - \frac{1}{\gamma u_2} \left[\frac{(\gamma+1)}{2} a^{*2} - \frac{(\gamma-1)}{2} u_2^2 \right]$$

Rearranging the terms of above equation, we get

$$u_2 - u_1 = \frac{1}{\gamma u_1} \frac{(\gamma+1)}{2} a^{*2} - \frac{1}{\gamma u_1} \frac{(\gamma-1)}{2} u_1^2 - \frac{1}{\gamma u_2} \frac{(\gamma+1)}{2} a^{*2} + \frac{1}{\gamma u_2} \frac{(\gamma-1)}{2} u_2^2$$

$$u_2 - u_1 = \frac{1}{\gamma u_1} \frac{(\gamma+1)}{2} a^{*2} - \frac{1}{\gamma u_2} \frac{(\gamma+1)}{2} a^{*2} + \frac{1}{\gamma u_2} \frac{(\gamma-1)}{2} u_2^2 - \frac{1}{\gamma u_1} \frac{(\gamma-1)}{2} u_1^2$$

Further rearrangements gives

$$u_2 - u_1 = \frac{(\gamma+1)}{2\gamma} a^{*2} \left[\frac{u_2 - u_1}{u_1 u_2} \right] + \frac{(\gamma-1)}{2\gamma} [u_2 - u_1]$$

$$u_2 - u_1 = \frac{(\gamma+1)}{2\gamma} a^{*2} \left[\frac{1}{u_1} - \frac{1}{u_2} \right] + \frac{(\gamma-1)}{2\gamma} [u_2 - u_1]$$

$$u_2 - u_1 = \frac{(\gamma+1)}{2\gamma} \left[\frac{a^{*2}}{u_1 u_2} \right] + \frac{(\gamma-1)}{2\gamma}$$

Necessary rearrangement for the above equation is as given,

$$\left[\frac{a^{*2}(\gamma+1)}{u_1 u_2} \right] \frac{1}{2\gamma} = \frac{2\gamma - \gamma + 1}{2\gamma}$$

$$a^{*2} = u_1 u_2$$

$$\frac{u_1}{a^{*2}} = \frac{u_2}{a^{*2}} \Rightarrow M_1^{*2} = \frac{1}{M_2^{*2}}$$

This expression shows that, M_1^{*2} and M_2^{*2} are reciprocal of each other for a normal shock. This equation is called as Prandtl's relation for normal shock which can be used to prove that Mach number becomes subsonic behind the normal shock

7.1 Deivation for Mach number based on starred quantities:

The Prandtl's expression gives the relation between the the starred acoustic speed based Mach numbers ahead and behind the shock. Using this expression, we can prove that $M_1^{*2} > 1$ which leads to $M_2^{*2} < 1$. Let's derive the expression initially for for M^*

$$\therefore M^{*2} = \frac{(\gamma+1)M^2}{2+(\gamma-1)M^2}$$

Which clearly proves that, for $M > 1$, $M^* > 1$ for $M = 1$, $M^* = 1$ and $M < 1$, $M^* < 1$

Therefore, from Prandtl's relation

$$\frac{1}{M_1^{*2}} = M_2^{*2} \quad \text{where } M_1 > 1 \Rightarrow M_1^{*2} > 1 \text{ \& } M_2 < 1 \Rightarrow M_2^{*2} < 1$$

This clearly proves that, any supersonic flow while pass through normal shock attains subsonic speed behind it.

From known freestream Mach number or the Mach number ahead of the shock we can calculate the Mach number behind the shock using Prandtl's relation.

$$\frac{1}{M_1^2} = M_2^2$$

$$\therefore \frac{(\gamma+1)M_1^2}{2+(\gamma-1)M_1^2} = \frac{1}{\frac{(\gamma+1)M_2^2}{2+(\gamma+1)M_2^2}}$$

$$\therefore M_2^2 = \frac{1 + \left[\frac{(\gamma+1)}{2} \right] M_1^2}{\gamma M_1^2 - \frac{(\gamma-1)}{2}}$$

We can derive the expression for the properties behind the shock wave using 1D conservation equations and know properties ahead the shock.

$$\rho_1 u_1 = \rho_2 u_2 \quad \text{Mass Conservation Equation}$$

$$\therefore \rho_1 u_1^2 = \rho_2 u_1 u_2 \quad \text{Multiplying both the sides by } u_1$$

$$\therefore \rho_1 u_1^2 = \rho_2 a^2 \quad \text{Using Prandtl's relation}$$

$$\frac{\rho_2}{\rho_1} = \frac{u_1^2}{a^2} = M_1^2$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_1^2}{2+(\gamma-1)M_1^2} \quad \text{Using relation between } M_1^2 \text{ \& } M_1^2$$

This equation gives the density ratio which is function of freestream Mach number and the specific heat ratio. We can find out the velocity ratio from this density ratio as,

$$\rho_1 u_1 = \rho_2 u_2 \quad \text{Mass Conservation Equation}$$

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2}$$

$$\frac{u_1}{u_2} = \frac{(\gamma+1)M_1^2}{2+(\gamma-1)M_1^2}$$

Let's derive the expression for static pressure ratio. For simplicity of derivation, initially representation of dynamic pressure is necessary and can be expressed as follows

$$\rho_1 u_1^2 = \frac{\rho_1}{\gamma p_1} u_1^2 \gamma p_1 = \gamma p_1 \frac{u_1^2}{a_1^2} = \gamma p_1 M_1^2 \quad \text{From definition of Mach number} \quad M = \frac{u}{a}$$

We know the 1D momentum conservation equation as

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

Replacing the dynamic pressure from either side and rearranging we get

$$p_2 - p_1 = \gamma p_1 M_1^2 \left[1 - \frac{u_2}{u_1} \right]$$

$$\frac{p_2 - p_1}{p_1} = \gamma M_1^2 \left[1 - \frac{u_2}{u_1} \right]$$

$$\frac{p_2}{p_1} - 1 = \gamma M_1^2 \left[1 - \frac{u_2}{u_1} \right]$$

$$\text{But } \frac{u_1}{u_2} = \frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_1^2}{2 + (\gamma-1)M_1^2}$$

$$\therefore \frac{p_2}{p_1} - 1 = \gamma M_1^2 \left[1 - \frac{2 + (\gamma-1)M_1^2}{(\gamma+1)M_1^2} \right]$$

$$\therefore \frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1)$$

This equation gives the static pressure ratio which is again function of freestream Mach number and the specific heat ratio. Temperature ratio can be obtained from the pressure and density ratio as,

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right) \left(\frac{\rho_2}{\rho_1} \right)$$

$$\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right] \left[\frac{2 + (\gamma-1)M_1^2}{(\gamma+1)M_1^2} \right]$$

Expression for static enthalpy ratio across the shock will be same as the static temperature ratio given by (7.4).

Following things are clear from the property ratios,

- All the ratios are dependant on freestream Mach number and specific heat ratio (γ), where Mach number is flow property and specific heat ratio is fluid property.
- Static pressure ratio, $\frac{P_2}{P_1}$, increases with increase in γ for same freestream Mach number. Hence, $\frac{P_2}{P_1}$ will be more for air than carbon-dioxide for the same Mach number since γ for carbon-dioxide is 1.28 while for air is 1.4. The variation in γ can occur for the same gas due to dissociation, ionization or vibrational excitation, combustion, mixing of two gasses of different specific heat ratios etc.
- $\frac{P_2}{P_1}$ increases with increase in Mach number for same γ (for same gas).
- $\frac{\rho_2}{\rho_1}$ ratio increases with increase in Mach number for same γ (for same gas).
- Density ratio, $\frac{\rho_2}{\rho_1}$, increases with decrease in γ , so for carbon-dioxide although it has lower $\frac{P_2}{P_1}$ for a given Mach number and γ , it will have higher $\frac{\rho_2}{\rho_1}$.
- $\frac{T_2}{T_1}$ has the same behavior as $\frac{P_2}{P_1}$ where it increases with increase in M and γ .
- Each $\frac{P_2}{P_1}$ and $\frac{T_2}{T_1} \rightarrow \infty$ as $M \rightarrow \infty$. However $\frac{\rho_2}{\rho_1}$ attains a constant value, $\frac{\gamma+1}{\gamma-1}$, as $M \rightarrow \infty$. For air ($\gamma = 1.4$) that constant value is 6.
- Mach number behind shock decreases as freestream Mach number increases.
However, as $M_1 \rightarrow \infty$, M_2 attains a constant value $\sqrt{\frac{\gamma-1}{2\gamma}}$ which is 0.378 for air ($\gamma = 1.4$)
- Mach number behind the normal shock (M_2) increases as γ increases for a given freestream Mach number.
- We know that entropy change in any process can be estimated as,

$$S_2 - S_1 = C_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$$

We can express the static temperature and pressure ratios in terms of freestream Mach number, Hence,

$$S_2 - S_1 = C_p \ln \left\{ \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right] \left[\frac{2 + (\gamma-1)M_1^2}{(\gamma+1)M_1^2} \right] \right\} - R \ln \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right]$$

From this expression it is clear that the entropy change would be positive if and only if $M_1 > 1$. That means shock is present only for supersonic flows and not for subsonic flows according to second law of thermodynamics.

- We know that 1D energy conservation equation is $h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$ If we image that the gas is isentropically brought to zero velocity on either side of the shock then the enthalpies would be stagnation enthalpies.

$$\therefore h_{01} = h_{02}$$

$$\therefore C_p T_{01} = C_p T_{02}$$

$$\therefore T_{01} = T_{02}$$

This expression re-asserts that flowfield is adiabatic since total temperature of the gas remains constant across the shock. Hence compression through shock is adiabatic irreversible.

- We can use already derived isentropic relation of total density to static density, eqto obtain the total density ratio across the shock.
-

$$\frac{\frac{\rho_{o1}}{\rho_1}}{\frac{\rho_{o2}}{\rho_2}} = \frac{\rho_{o1}}{\rho_{o2}} \frac{\rho_2}{\rho_1} = \frac{\left[1 + \frac{\gamma-1}{2} M_1^2\right]^{\frac{1}{\gamma-1}}}{\left[1 + \frac{\gamma-1}{2} M_2^2\right]^{\frac{1}{\gamma-1}}}$$

$$\therefore \frac{\rho_{o2}}{\rho_{o1}} = \frac{\left[1 + \frac{\gamma-1}{2} M_2^2\right]^{\frac{1}{\gamma-1}}}{\left[1 + \frac{\gamma-1}{2} M_1^2\right]^{\frac{1}{\gamma-1}}} \frac{\rho_2}{\rho_1}$$

$$\therefore \frac{\rho_{o2}}{\rho_{o1}} = \left[\frac{1 + \frac{\gamma-1}{2} M_2^2}{1 + \frac{\gamma-1}{2} M_1^2} \right]^{\frac{1}{\gamma-1}} \left[\frac{(\gamma+1) M_1^2}{2 + (\gamma-1) M_1^2} \right]$$

•

Similarly we can find out the total pressure ratio across the shock using the isentropic relation given by eq. (6.2).

$$\frac{\frac{P_{o2}}{P_2}}{\frac{P_{o1}}{P_1}} = \frac{P_{o2}}{P_2} \frac{P_2}{P_1} = \frac{\left[1 + \frac{\gamma-1}{2} M_1^2\right]^{\frac{\gamma}{\gamma-1}}}{\left[1 + \frac{\gamma-1}{2} M_2^2\right]^{\frac{\gamma}{\gamma-1}}} \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1)\right]$$

•

Total pressure and total density ratios are of same magnitude for the ideal gas conditions, since total temperature remains constant across the shock

This can be proved using ideal gas equation.

$$P = \rho RT$$

$$\therefore \frac{P}{\rho} = RT$$

$$\therefore \frac{P_{o1}}{\rho_{o1}} = RT_{o1}$$

$$\therefore \frac{P_{o2}}{\rho_{o2}} = RT_{o2}$$

$$\therefore \frac{P_{o1}}{\rho_{o1}} = \frac{P_{o2}}{\rho_{o2}}$$

$$\therefore \frac{P_{o2}}{P_{o1}} = \frac{\rho_{o2}}{\rho_{o1}}$$

- We have derived all the static and total property ratios as function of M1 & γ , however, we can always express these relations as function of M2 & γ , since M2 is also function M1.
- We can calculate the entropy change across the shock given by eq. using the presently derived ratios,

$$S_2 - S_1 = C_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$$

- If we imagine that the gas element is isentropically brought to zero on either sides of shock then

$$S_1 = S_{o1}, S_2 = S_{o2}$$

$$\therefore S_2 - S_1 = S_{o2} - S_{o1}$$

$$\therefore S_2 - S_1 = C_p \ln \frac{T_{o2}}{T_{o1}} - R \ln \frac{P_{o2}}{P_{o1}}$$

$$\therefore S_2 - S_1 = -R \ln \frac{P_{o2}}{P_{o1}}$$

$$\frac{P_{o1}}{P_{o2}} = e^{-\frac{\Delta S}{R}}$$

This proves that $\frac{P_{o2}}{P_{o1}}$ is less than 1 since ΔS is positive for shocked flow. This expression suggests that the total pressure decreases across the normal shock.

Hugoniot equation

Shock wave compression is an adiabatic and irreversible process; hence shock waves can be treated as irreversible adiabatic compressors. Let's calculate the work required for this compression process.

$$\rho_1 u_1 = \rho_2 u_2$$

$$u_1 = \left(\frac{\rho_2}{\rho_1} \right) u_2$$

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

$$p_1 + \rho_1 \left(\frac{\rho_2}{\rho_1} \right)^2 u_2^2 = p_2 + \rho_2 u_2^2$$

$$p_1 + \frac{\rho_2^2 u_2^2}{\rho_1} = p_2 + \rho_2 u_2^2$$

$$u_2^2 \left(\frac{\rho_2^2}{\rho_1} - \rho_2 \right) = p_2 - p_1$$

$$u_2^2 \left(\frac{\rho_2^2 - \rho_1 \rho_2}{\rho_1} \right) = p_2 - p_1$$

$$u_2^2 \frac{\rho_2}{\rho_1} (\rho_2 - \rho_1) = p_2 - p_1$$

$$u_2^2 = \frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_1}{\rho_2} \right)$$

Similarly

$$u_1^2 = \frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_2}{\rho_1} \right)$$

We know from the energy equation that,

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \quad \text{Energy Conservation}$$

$$\frac{p_1}{\rho_1} + e_1 + \frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_2}{\rho_1} \right) \frac{1}{2} = \frac{p_2}{\rho_2} + e_2 + \frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_1}{\rho_2} \right) \frac{1}{2}$$

Putting the expression for u_1 and u_2

$$\begin{aligned}
 e_2 - e_1 &= \frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} + \frac{1}{2} \frac{(p_2 - p_1)}{\rho_2 - \rho_1} \left(\frac{\rho_2}{\rho_1} \right) - \frac{1}{2} \frac{p_2 - p_1}{\rho_2 - \rho_1} \left(\frac{\rho_1}{\rho_2} \right) \\
 e_2 - e_1 &= \frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} + \frac{\rho_2^2 (p_2 - p_1) - \rho_1^2 (p_2 - p_1)}{2(\rho_2 - \rho_1) \rho_1 \rho_2} \\
 e_2 - e_1 &= \frac{p_1}{\rho_1} - \frac{p_2}{\rho_2} + \frac{p_1(\rho_1^2 - \rho_2^2) - p_2(\rho_1^2 - \rho_2^2)}{2(\rho_2 - \rho_1) \rho_1 \rho_2} \\
 e_2 - e_1 &= \frac{2(\rho_2 - \rho_1) \rho_2 p_1 - 2(\rho_2 - \rho_1) \rho_1 p_2 + p_1(\rho_1^2 - \rho_2^2) - p_2(\rho_1^2 - \rho_2^2)}{2(\rho_2 - \rho_1) \rho_1 \rho_2} \\
 e_2 - e_1 &= \frac{2(\rho_2 - \rho_1) \rho_2 p_1 - 2(\rho_2 - \rho_1) \rho_1 p_2 + p_1(\rho_1^2 - \rho_2^2) - p_2(\rho_1^2 - \rho_2^2)}{2(\rho_2 - \rho_1) \rho_1 \rho_2} \\
 e_2 - e_1 &= \frac{p_1 \rho_2^2 - 2p_1 \rho_1 \rho_2 + p_1 \rho_1^2 + p_2 \rho_2^2 - 2p_2 \rho_1 \rho_2 + p_2 \rho_1^2}{2(\rho_2 - \rho_1) \rho_1 \rho_2} \\
 e_2 - e_1 &= \frac{p_1(\rho_2 - \rho_1)^2 + p_2(\rho_2 - \rho_1)^2}{2(\rho_2 - \rho_1) \rho_1 \rho_2} \\
 e_2 - e_1 &= \left(\frac{p_2 + p_1}{2} \right) \left[\frac{(\rho_2 - \rho_1)}{\rho_1 \rho_2} \right] \\
 e_2 - e_1 &= \left(\frac{p_2 + p_1}{2} \right) \left[\frac{1}{\rho_1} - \frac{1}{\rho_2} \right]
 \end{aligned}$$

The above equation is called as the Hugoniot equation which relates thermodynamic quantities across the normal shock. This relation is same as the first law of thermodynamics which states that internal energy change for an adiabatic process is equal to work done or pressure (average pressure in the present case) times change in specific volume.

We can plot the Hugoniot curve for the known initial pressure-volume conditions for various pressure-volume conditions after normal shock. Hence Hugoniot curve, in principle, joins all the possible points on p-v plane starting from a known point on p-v plane. To generate such a plot we have to modify the Hugoniot equation by expressing the internal energy in terms of the pressure and volume as,

$$e = c_v T = \frac{R}{\gamma - 1} T = \frac{pv}{\gamma - 1}$$

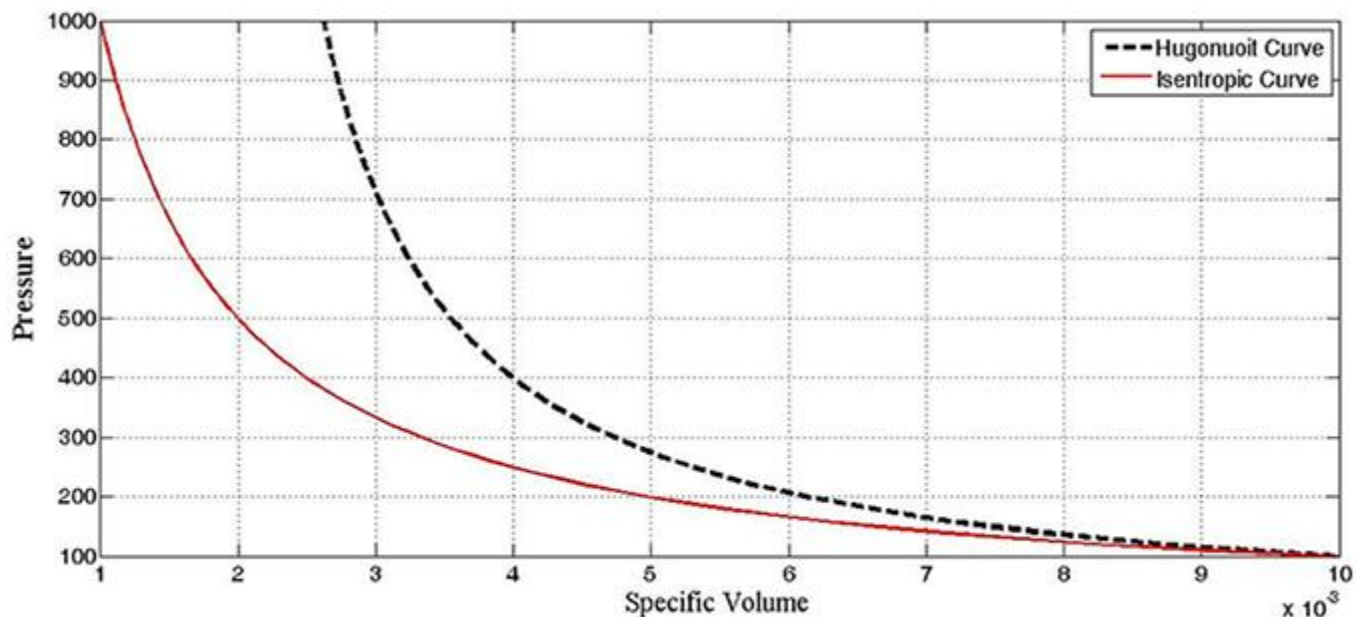
$$e_2 - e_1 = \left(\frac{p_2 + p_1}{2} \right) [v_1 - v_2]$$

$$\left(\frac{p_2 v_2 - p_1 v_1}{\gamma - 1} \right) = \left(\frac{p_2 + p_1}{2} \right) [v_1] - \left(\frac{p_2 + p_1}{2} \right) [v_2]$$

$$v_2 \left[\frac{p_2}{\gamma - 1} + \frac{p_2 + p_1}{2} \right] = v_1 \left[\frac{p_1}{\gamma - 1} + \frac{p_2 + p_1}{2} \right]$$

$$v_2 = v_1 \frac{\left[\frac{p_1}{\gamma - 1} + \frac{p_2 + p_1}{2} \right]}{\left[\frac{p_2}{\gamma - 1} + \frac{p_2 + p_1}{2} \right]}$$

Using this expression, we can get all the possible values of v_2 for given values P_1 , P_2 and v_1 and hence we can plot the p - v diagram or Hugoniot curve for normal shock. The line joining initial point and any point of the curve specifies particular massflow rate and hence a particular freestream Mach number.



Now let's try to understand the basics of normal shock in view of this Hugoniot relation. The expression for specific volume is a consolidated equation for normal

shock conditions. If we have a normal shock condition due to supersonic velocity u_1 , then as we know from eq. (8.2)

$$u_1^2 = \frac{P_2 - P_1}{\rho_2 - \rho_1} \left(\frac{\rho_2}{\rho_1} \right) \text{ Since } \rho = \frac{1}{v}$$

$$u_1^2 = \frac{P_2 - P_1}{\frac{1}{v_2} - \frac{1}{v_1}} \left(\frac{v_1}{v_2} \right)$$

$$\frac{P_2 - P_1}{v_2 - v_1} = - \left(\frac{u_1}{v_1} \right)^2$$

This expression gives the slope of a straight line joining two points of Hugoniot curve, mainly initial point and any other point on the Hugoniot curve on p-v diagram. These two points necessarily define the upstream and downstream locations of the normal shock respectively. Since this expression is for slope and is comprised of velocity and density (inverse of specific volume), this line necessarily corresponds to a particular mass flow rate.

$$\frac{P_2 - P_1}{v_2 - v_1} = - \left(\rho_1 u_1 \right)^2 = - \left(\text{Massflux} \right)^2$$

Therefore if we know the mass flow rate and initial conditions we can easily find out the post shock conditions. These conditions are given by point of intersection of Hugoniot curve and the straightline drawn from initial conditions with slope equation given by eq.

In the same figure Hugoniot curve is plotted along with isentropic curve for compression. These curves originate from the same point. Slope of curve representing isentropic compression can be calculated as,

$$pv^\gamma = \text{const}$$

$$\frac{dp}{dv} v^\gamma + p \gamma v^{\gamma-1} = 0$$

$$\frac{dp}{dv} = -\frac{p\gamma}{v} = -p\gamma\rho = -\frac{p\gamma\rho^2}{\rho} = -a^2 \rho^2$$

$$\frac{dp}{dv} = -\left(\frac{a^2}{v^2}\right)$$

This suggests that, at the initial condition is same for isentropic and Hugoniot curves. This means that the slope of both the curves is same at that point. This proves that the, we have $u_1 = a_1$, $M_1 = 1$ at the initial or starting point. Therefore for sonic flow there is no change in properties across the shock. As freestream Mach number increases u_1 becomes greater than a_1 and

slope $(\tan \alpha) = -\left[\frac{u_1}{v_1}\right]_{\text{Hugoniot}}^2 < -\left[\frac{a_1}{v_1}\right]_{\text{isentrop}}^2$ so $\left|\left[\frac{u_1}{v_1}\right]^2\right| > \left|\left[\frac{a_1}{v_1}\right]^2\right|$ Therefore slope of Hugoniot curve is more than the slope for isentropic curve for supersonic flows. Hence the graph herewith provides an evidence for the fact that, although isentropic compression is efficient, compression by normal shock is more effective since compression by normal shock gives higher pressure rise for same change in specific volume. Hence shock can also be called as thermodynamic compressor.

Hugoniot Curve

We have understood the typical Hugoniot curve in Figure 8.1. As per the Hugoniot curve, there are many possible states behind the shock for a given initial condition. Therefore to arrive at a particular state we will have know the massflux or Mach number. A straight line of the slope proportional to the massflux or Mach number when drawn along with Hugoniot curve, gives a fixed state behind the normal shock for known upstream state properties. The point at which this straight line cuts the Hugoniot curve corresponds to the state behind

the shock. Possibility of such a state is still unclear since the procedure for finding the post shock condition is based on Hugoniot relation which in turn is based on mass, momentum equations and first law of thermodynamics. The straight line equation is also based on mass and momentum equations only. Therefore to select a possible state for a given initial conditions and known massflux, we have to use second law of thermodynamics. Here we can prove the impossibility of expansion shock and presence of shock in supersonic and hypersonic flows.

We know that the Hugoniot equation is

$$e_2 - e_1 = u_2 - u_1 = \frac{p_1 + p_2}{2} [v_1 - v_2]$$

Since we know properties at station 1 therefore these can be treated as constants in this equation.

$$e_2 - e_1 = u_2 - u_1 = \frac{p_1 + p_2}{2} v_1 - \frac{p_1 + p_2}{2} v_2$$

Differentiating above equation

$$du = \frac{p_1 + p_2}{2} (0 - dv) + \frac{v_1 - v_2}{2} (0 + dp)$$

$$du = \frac{p_1 + p_2}{2} (-dv) + \frac{v_1 - v_2}{2} (dp)$$

We can assume that there is a reversible heat addition process in a closed system for which we will have same initial and final states as that of the normal shock. Therefore the internal energy and entropy change for assumed process will be same as that of the normal shock case. Lets apply laws of thermodynamics to the assumed process.

$$dQ = du + p dv$$

$$T ds = du + p dv$$

$$du = T ds - p dv$$

Since change in internal energy is same for both the processes, let's equate the change in internal energy after removing subscript 2 we get,

$$T ds - p dv = \frac{p_1 + p}{2} (-dv) + \frac{v_1 - v}{2} (dp)$$

$$T ds - p dv = -\frac{1}{2} p_1 dv - \frac{1}{2} p dv + \frac{1}{2} v_1 dp - \frac{1}{2} v dp$$

$$T ds = -\frac{1}{2} p_1 dv + \frac{1}{2} p dv + \frac{1}{2} v_1 dp - \frac{1}{2} v dp$$

$$T ds = \frac{1}{2} dv (p - p_1) - \frac{1}{2} dp (v - v_1)$$

$$2T ds = dv (p - p_1) - dp (v - v_1)$$

$$2T \frac{ds}{dp} = (v_1 - v) \left[1 - \frac{(p_1 - p)}{v_1 - v} \frac{dv}{dp} \right]$$

Here T is the temperature at which heat is added reversibly to the assumed process. But we are overlapping the normal shock with the assumed reversible heat addition process. However, from combination of mass and momentum equation we have already derived the slope for normal shock in terms of massflow rate. The same thing is mentioned herewith in short.

$$\text{Mass flux} = \rho v = \rho_1 v_1 = \rho_2 v_2 = \text{constant} = k$$

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

$$p_1 + \frac{k^2}{\rho_1} = p_2 + \frac{k^2}{\rho_2}$$

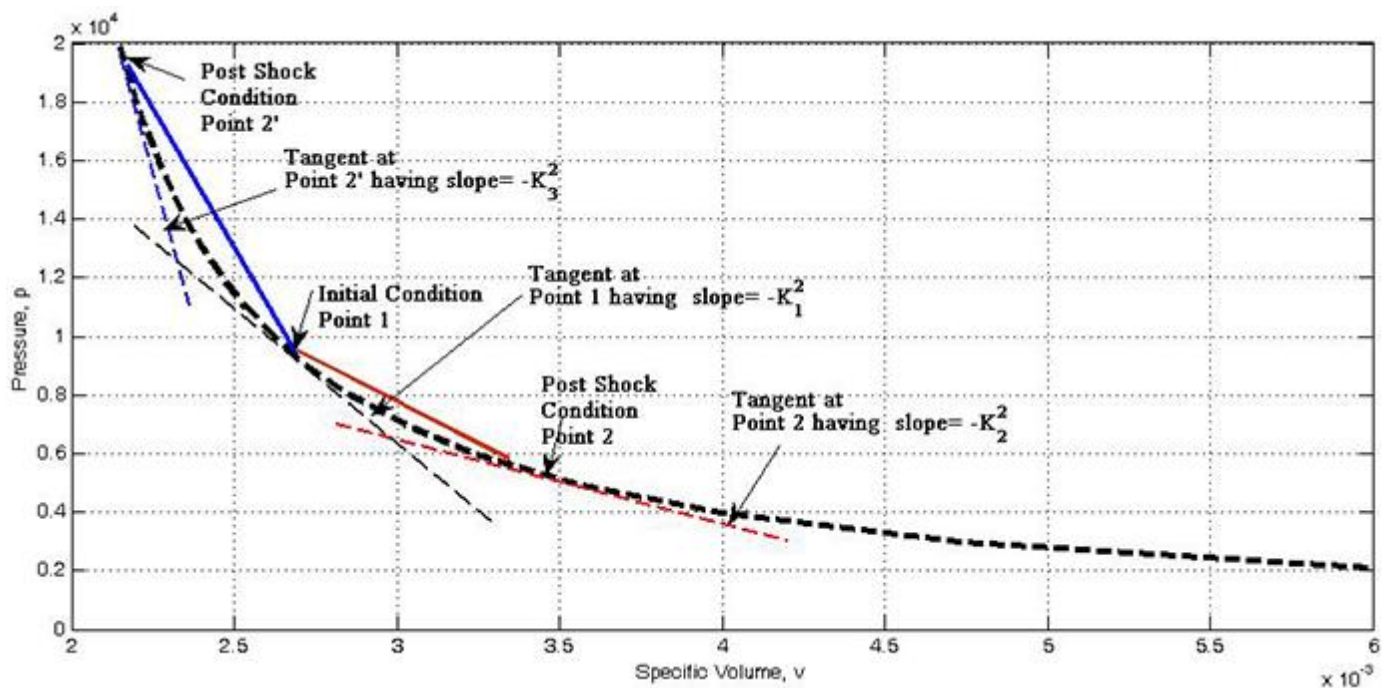
$$p_1 + v_1 k^2 = p_2 + v_2 k^2$$

$$k^2 = -\frac{p_1 - p_2}{v_1 - v_2}$$

This is the slope of the straight line drawn to intersect Hugoniot curve. Using above expression, we can re-write equation (9.1) as,

$$2T \frac{ds}{dp} = (v_1 - v) \left[1 + \frac{k^2}{dp/dv} \right]$$

Since derivatives in the above expression are of the properties of system which can belong to any process (normal shock or reversible heat addition).



Analysis of normal shock using Hugoniot Curve

The above figure will help us in finding most probable post shock condition. In this figure, along with Hugoniot curve, three different lines are drawn to get various possible post shock conditions. These lines correspond to different massflow rates or freestream conditions forming the normal shock. One of those lines is tangent to Hugoniot curve at point 1 which represents the initial condition or pre-shock conditions. Hence the straight line at point 1 has slope $(-k_1^2)$ which is same as the Hugoniot curve at that point, $\frac{dp}{dv}$. Therefore term in bracket of

eq. (9.3), $\left[1 + \frac{k^2}{\frac{dp}{dv}}\right]$ becomes 2. However, $(v_1 - v)$ is equal to zero since there is no change in specific volume. Therefore $\frac{ds}{dp}$ is equal to zero in turn dS is equal to zero which makes this process possible.

Now consider the line with slope k_2^2 intersecting at point 2 to Hugoniot curve. For this case right hand side of eq. (9.3) can be written as,

$$\left[1 + \frac{k^2}{\frac{dp}{dv}}\right] = 1 - \frac{-k_2^2}{\left[\frac{dp}{dv}\right]_2} = 1 - \frac{\text{Slope of straightline joining 1\&2}}{\text{Slope of tangent at point 2}}$$

Equation (9.3) for the corresponding line can be written as,

$$2T \left(\frac{\partial s}{\partial p} \right)_{at 2} = (v_1 - v_2) \left[1 + \frac{k_2^2}{\left(\frac{\partial p}{\partial v} \right)_{at 2}} \right]$$

Normal Shock analysis using Hugoniot Curve

Now, we have to find out whether point 2 is possible state or not. Here point 2 is a representative point for all points on Hugoniot curve below point 1 or for which specific volume increases with decrease in pressure, since slope of the straight line joining initial and final state, for all these points, is less than the slope of Hugoniot curve at the initial state. Hence equation provides a negative number. But $(v_1 - v)$ is also a negative number since specific volume is more at state 2

than at state 1. Therefore $\frac{ds}{dp}$ is positive from Equation. However since pressure decreases from state 1 to state 2 dp is negative. Therefore dS has to be negative for negative $\frac{ds}{dp}$. This is violation of second law of thermodynamics since entropy of system is decreasing. Hence representative point 2 is not a possible post

shock state where pressure decreases and specific volume increases across the shock. Therefore second law of thermodynamics denies possibility of expansion shock. Therefore any post shock state for which slope of straight line joining initial state and final state is more than the slope of the Hugoniot curve at final state is not a possible post shock state.

Now consider any point (2') above point 1 as a post shock state for which pressure is more and specific volume is less. Lets join point 1 and 2' by a straight line and draw a tangent to Hugoniot curve at 2'. Term in bracket of eq. (9.3) can be this special case as,

$$2T \left(\frac{\partial s}{\partial p} \right)_{2'} = (v_1 - v_{2'}) \left[1 - \frac{-k_{2'}^2}{\left(\frac{\partial p}{\partial v} \right)_{2'}} \right]$$

Therefore,

$$\left[1 + \frac{k^2}{\frac{dp}{dv}} \right] = 1 - \frac{-k_{2'}^2}{\left[\frac{dp}{dv} \right]_{2'}} = 1 - \frac{\text{Slope of straightline joining 1\&2'}}{\text{Slope of tangent at point 2'}}$$

As we know $\left| (-k_{2'}^2) \right| < \left| \left(\frac{\partial p}{\partial v} \right)_{2'} \right|$ so term in bracket of Equation is above zero and below 1 and indeed positive. Also the term $v_1 - v_2$ is a positive, since $v_{2'} < v_1$.

Hence, from equation (10.2) $\left(\frac{\partial s}{\partial p} \right)_{2'} > 0$. However pressure increases in this

process, hence entropy has to increase to keep $\left(\frac{\partial s}{\partial p} \right)_{2'} > 0$. Therefore for all points on Hugoniot curve having slope more than that of the slope of at the initial condition represent the possible post shock condition. This means that the massflux line should have at least slope equal to the tangent drawn to Hugoniot curve at initial state where initial state is the post shock state. Mathematical representation of this statement is

Slope of mass fluxline or line joining initial and final states = Slope of tangent to Hugoniot curve at initial state

$$\left[1 + \frac{k^2}{dp/dv}\right] = 1 - \frac{-k_2'^2}{\left[dp/dv\right]_{2'}} = 1 - \frac{\text{Slope of straightline joining 1\&2'}}{\text{Slope of tangent at point 2'}}$$

Because $\frac{\partial p}{\partial v} = \frac{\partial p}{\partial \rho} \frac{\partial \rho}{\partial v}$ and $\frac{\partial p}{\partial v} = -\frac{\rho}{v} = -\rho^2$

But, we know from eq that $K_1 = \rho_1 V_1$. Putting this in equation we get,

$$\rho_1 V_1^2 = \rho_1 \frac{dp}{d\rho}$$

$$V_1^2 = \frac{dp}{d\rho}$$

As we know that $\frac{dp}{d\rho} = a^2$

$$V_1^2 = a^2$$

$$M_1 = 1$$

Herewith we have seen that, $M = 1$ is the limiting case for which post shock state is thermodynamically possible. However change in pressure and specific volume for this case is zero and the change in entropy as well. Therefore, shock for Mach 1 flow is called as Mach line. We have thus proved that if slope of massfluxline or straight line joining initial and final states is less than the slope of tangent to Hugoniot curve at final state then there is no shock possible since massflow is less than the critical value. However since density is same for both the cases, lesser massflux indicates lower speed. But speed for the critical case itself is sonic hence straight line joining initial state and representative point 2 represents subsonic speed. This proves the fact that subsonic flow will not encounter any shock.

Differential Equations of Normal Shock

Hugoniot relation suggests that shock can happen only for supersonic flows and Prandtl's relation proves that Mach number decreases across the normal shock. Now we are interested to prove what the qualitative variation for other properties of flow across normal shock.

$$M = \frac{V}{a} = \frac{V}{\sqrt{\gamma RT}}$$

$$dM = \frac{1}{a} dV - \frac{V}{a^2} da$$

Since,

$$a = \sqrt{\gamma RT}$$

$$da = \sqrt{\gamma R} \cdot \frac{dT}{2\sqrt{T}}$$

$$\frac{da}{a} = \frac{\sqrt{\gamma R}}{\sqrt{\gamma RT}} \cdot \frac{dT}{2\sqrt{T}} = \frac{dT}{2T}$$

Therefore,

$$dM = M \cdot \frac{dV}{V} - M \cdot \frac{da}{a}$$

$$\frac{dM}{M} = \frac{dV}{V} - \frac{da}{a} = \frac{dV}{V} - \frac{dT}{2T}$$

However

$$h + \frac{V^2}{2} = \text{const}$$

$$dh + V dV = 0 \Rightarrow \frac{dh}{h} = -\frac{V dV}{C_p T} = -\frac{V dV}{\gamma R T / (\gamma - 1)}$$

$$\frac{dT}{T} = \frac{dh}{h} = -\frac{V dV}{a^2 / (\gamma - 1)} = -(\gamma - 1) M^2 \frac{dV}{V}$$

Hence,

$$\frac{dM}{M} = \frac{dV}{V} - \frac{dT}{2T} = \frac{dV}{V} + \frac{1}{2}(\gamma - 1) M^2 \frac{dV}{V}$$

$$\frac{dM}{M} = \frac{dV}{V} \left[1 + \frac{1}{2}(\gamma - 1) M^2 \right]$$

$$\frac{dV}{V} = \frac{dM}{M} \cdot \frac{1}{\left[1 + \frac{1}{2}(\gamma - 1) M^2 \right]}$$

Since the term dM on the right hand side is negative for normal shock due to decrease of Mach number across it, we will get dV also negative. This proves

that velocity decreases across the shock. But $\frac{dT}{T} = -(\gamma - 1) M^2 \frac{dV}{V}$ which proves dT is positive and temperature increases across the shock.

Now let's consider 1D momentum conservation equation,

$$p + \rho V^2 = \text{const}$$

$$dp + 2\rho V dV + V^2 d\rho = 0$$

$$\text{but } pV = \text{const, Therefore } \frac{dp}{\rho} = -\frac{dV}{V}$$

$$dp + 2\rho V dV + V^2 \left(-\frac{\rho dV}{V} \right) = 0$$

$$dp + 2\rho V dV + (-\rho V dV) = 0$$

$$dp + \rho V dV = 0$$

$$dp = -\rho V dV \Rightarrow \frac{dp}{\rho} = -\frac{\rho V dV}{\rho} = -\frac{\gamma \rho V^2}{\gamma \rho} \frac{dV}{V} = -\gamma M^2 \frac{dV}{V}$$

Since velocity decreases across the normal shock, static pressure increases across normal shock.

$$p = \rho R T$$

$$\frac{dp}{\rho} = \frac{dp}{\rho} + \frac{dT}{T}$$

$$\frac{dp}{\rho} = \frac{dp}{\rho} - \frac{dT}{T}$$

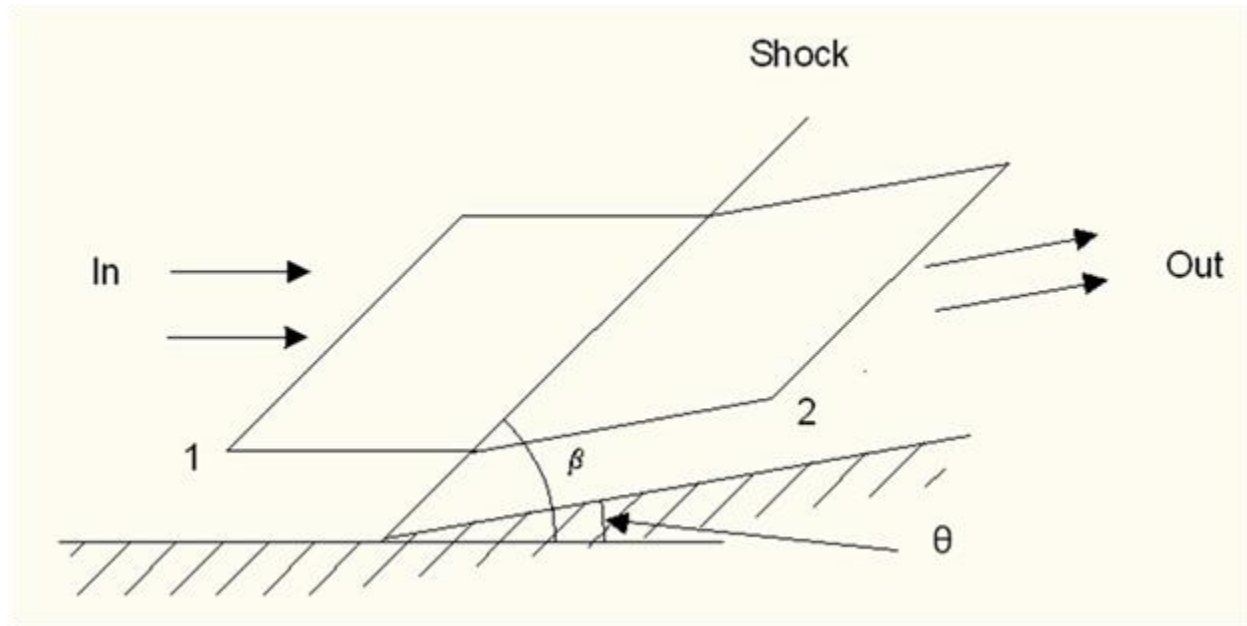
Both pressure and temperature increase across normal shock so we cannot

conclude from this relation. However, we can easily interpret from $\frac{dp}{\rho} = -\frac{dV}{V}$ that density increases across the normal shock.

This clearly demonstrates that percentage increase in pressure across normal shock is more than the percentage increase in temperature due to which density increases across the normal shock to keep mass flow rate same since velocity decreases across normal shock.

Oblique Shock

Consider the flow taking place along a wedge as shown in **Fig.** Let θ be the wedge angle and β be the shock angle with the wall which is parallel to the approaching freestream.



An oblique shock for a supersonic flow over the wedge.

As we have already proved that shock exists only for supersonic flows, consider a supersonic flow of Mach number M_1 approaching the wedge. In the presence of the shock, flow deflects by an angle θ which is the wedge angle. Let's solve the mass, momentum and energy equations for this flow. Consider the control volume as shown in **Fig.** In this special control volume, inlet and outlet are parallel to the shock. Other two faces of the control volume are parallel to the streamline hence these faces will not contribute to the mass, momentum and energy fluxes. Let u be the velocity normal to the shock and w be the velocity parallel to the shock. Graphical demonstration of these velocities is given in **Fig.** Station 1 corresponds to inlet or preshock conditions while station 2 corresponds to outlet or post shock conditions.

$$\frac{d}{dt} \int_V \rho dv + \oint_S \rho V \cdot ds = 0$$

Mass conservation in integral form

Let's assume the flow to be steady, hence,

$$\oint_S \rho V \cdot d\vec{s} = 0$$

$$(\rho V \cdot d\vec{s})_{post} - (\rho V \cdot d\vec{s})_{pre} = 0$$

$$(\rho V \cdot d\vec{s})_{post} = (\rho V \cdot d\vec{s})_{pre}$$

But

$$(V \cdot d\vec{s})_{post} = u_2 A \quad \text{and} \quad (V \cdot d\vec{s})_{pre} = u_1 A$$

Hence,

$$(\rho u)_2 = (\rho u)_1 \quad \text{or} \quad \rho_1 u_1 = \rho_2 u_2$$

This is the mass conservation equation for oblique shock conditions expressed in terms of velocities normal to the shock.

Now consider the momentum conservation equation for the same flow. Since momentum is the vector equation, we have to consider, two equations, viz, normal and parallel to the shock. Let's initially consider the momentum equation in integral form for inviscid flow.

$$\frac{\partial}{\partial t} \oint_V \rho \vec{V} dV + \oint_S (\rho \vec{V} \cdot d\vec{S}) \vec{V} = - \oint_S P d\vec{S}$$

For steady flow, this equation becomes,

$$\oint_S (\rho \vec{V} \cdot d\vec{S}) \vec{V} = - \oint_S P d\vec{S}$$

Now consider the momentum equation in the direction parallel to the shock wave. Since there is no pressure difference in this direction, the right hand side will be zero. Hence,

$$\oint_S (\rho \vec{V} \cdot d\vec{S}) w = 0$$

$$(\rho u w)_{pre} = (\rho u w)_{post}$$

but using mass conservation Eq we can re-write it as,

$$(w)_{\text{pre}} = (w)_{\text{post}} \text{ or } w_1 = w_2$$

This expression clearly suggests that velocity parallel to the shock remains conserved.

Now consider the momentum equation normal to the shock.

$$\oint_S (\rho V \cdot ds) u = - \oint_S (p \cdot ds)$$

$$\oint_S (\rho u u + p) \cdot ds = 0$$

$$\oint_S (p + \rho u^2) \cdot ds = 0$$

$$(p + \rho u^2)_{\text{post}} = (p + \rho u^2)_{\text{pre}}$$

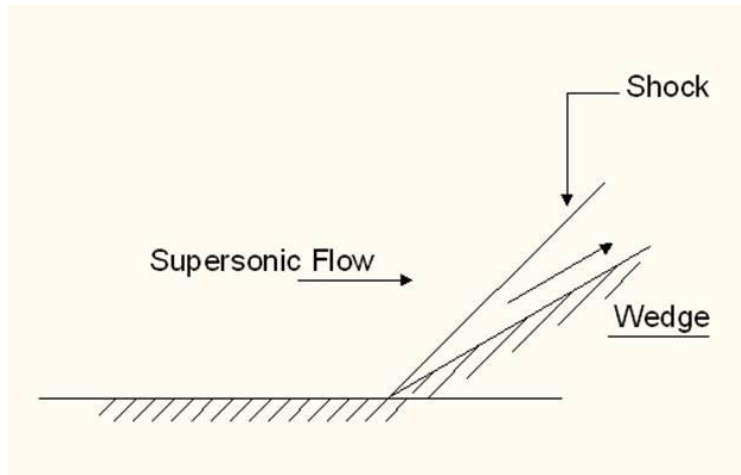
or

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

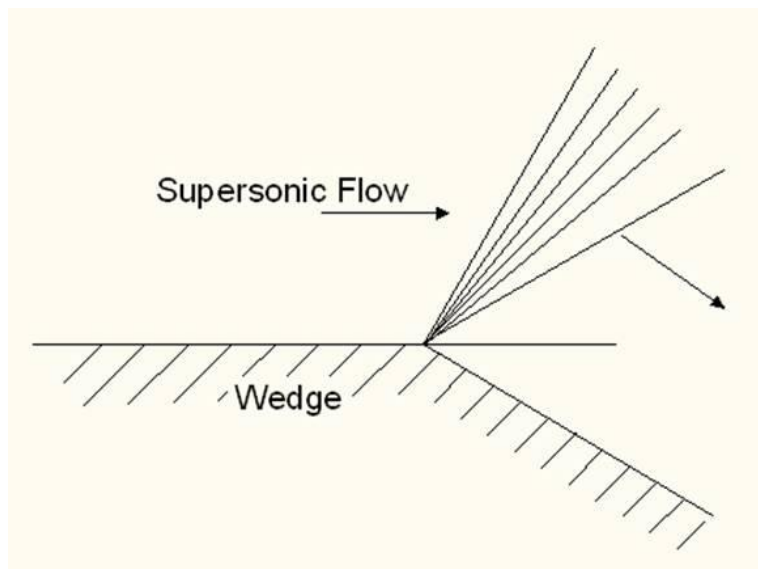
We can clearly see that the momentum equation looks exactly same as that for the normal shock relations. Here u is the velocity normal to the shock. Therefore only velocity component normal to the shock wave is responsible for the change in momentum since momentum and velocity tangential to shock are conserved.

The shock encountered by such the flow is the normal shock. Herewith we are going to deal the compressible flows in two dimensions, 2D. There are two entities which are going to tackle the expansion and compression of the 2D compressible flow. Such situations are two situation shown in **Fig.** which the flow might encounter. If the flow turns in to itself then the compression of flow takes place through shock as shown in **Fig.** Since this shock makes certain angle with the flow, it is called as the oblique shock. However if the flow turns away from itself then the expansion of the flow takes place through an expansion fan as shown in **Fig.** Deriving various relations for angles or properties ratios for the

two dimensional compressible flows in the presence of shock or expansion fan is the central objective here onwards.



Supersonic flow turning into itself in the presence of shock



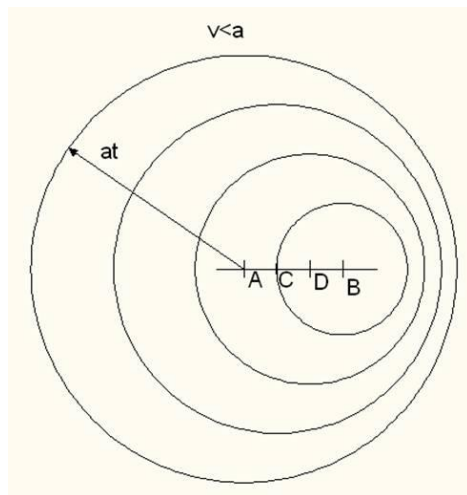
Supersonic flow turning away from itself in the presence of expansion fan

The main reason of having shock or expansion fan at the deflection corners is to deflect the flow accordingly so as to maintain its parallelism with the wall in order to avoid the direct collision. Therefore the angle of shock with the wall or

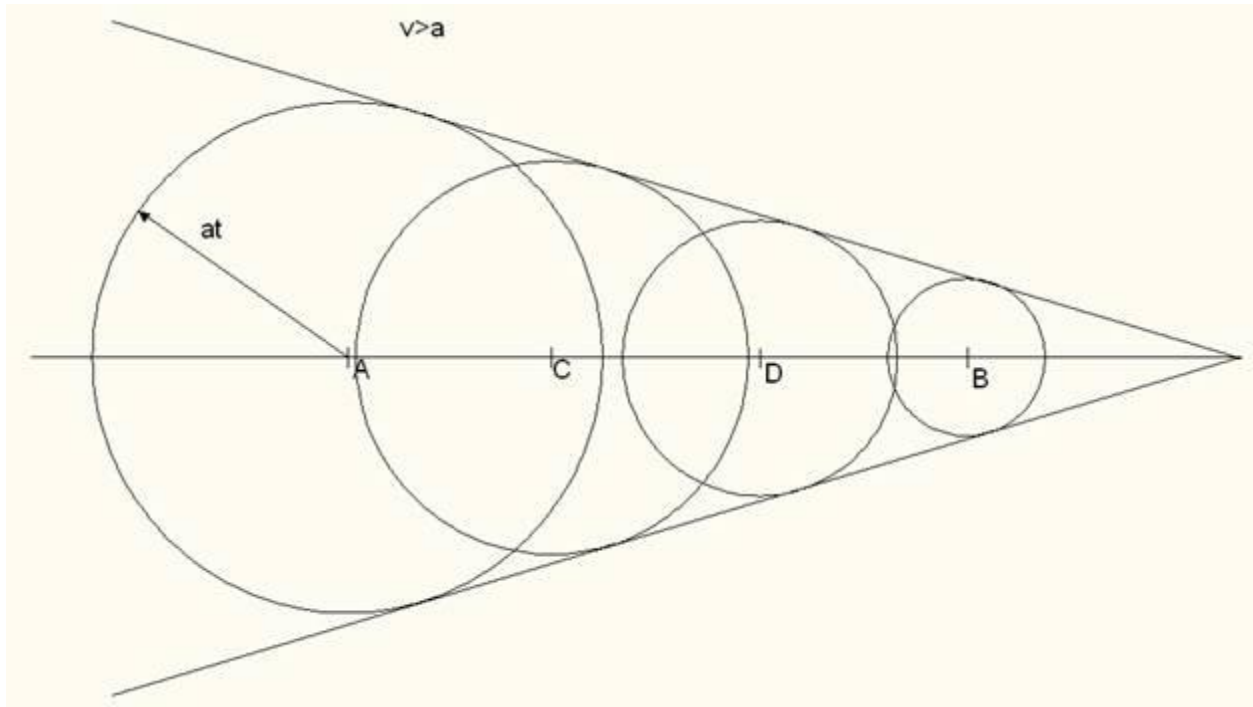
freestream velocity will be dependant on flow deflection angle and freestream velocity.

Mach Wave

Lets consider a raliway traveling along its track from location A to location B. If it makes a beep sound while leaving station A and at intermediate locations C and D then the sound wave travels spherically out with the velocity (say) 'a' m/s. The distance traveled by the sound waves originated from A,C and D locations are given by the circles drawn, considering respective points as centers. Now if the speed of the train is less than acustic velocity (a), say v m/s, then the train will always remain in side the circles. This statement also means that the distance travels by the train in some time interval is always lower than the distance traveled by the acustic wave in the same time interval. Now if we consider the speed of train to be more than speed of sound then the train will always cross the respective circle of acustic speed in that time interval. Now if we draw the tangent to all circles, representing position of acustic wave, from the end location of train (B), then we can see in **Fig.** that such a tangent makes an angle μ with the train track. The expression for this angle is as,



Train travelling from A to B at subsonic speed



Train travelling from A to B at supersonic speed

$$\sin(\mu) = \frac{at}{vt}$$

$$\sin(\mu) = \frac{1}{M}$$

$$\mu = \sin^{-1} \frac{1}{M}$$

This angle (μ) is called as Mach angle and it becomes the reference angle for our following discussions. Shock angle which we are going to understand is always more than the Mach angle

Oblique Shock

We have already derived the mass and momentum conservation equations for the oblique shock conditions. Consider the integral form of energy equation for

inviscid compressible flow.

$$\oint_V \frac{\partial}{\partial t} \left[\rho \left(e + \frac{V^2}{2} \right) \right] dV + \oint_S \rho \left(e + \frac{V^2}{2} \right) \vec{V} \cdot d\vec{S} = - \oint_S P \vec{V} \cdot d\vec{S}$$

For steady flow, this equation changes to,

$$\oint_S \rho \left(e + \frac{V^2}{2} \right) \vec{V} \cdot d\vec{S} = - \oint_S P \vec{V} \cdot d\vec{S}$$

$$\oint_S \rho \left(e + \frac{V^2}{2} \right) \vec{V} \cdot d\vec{S} + \oint_S P \vec{V} \cdot d\vec{S} = 0$$

$$\oint_S \left(\rho e + \frac{\rho V^2}{2} + p \right) \vec{V} \cdot d\vec{S} = 0$$

$$\left[\left(e + \frac{V^2}{2} + \frac{p}{\rho} \right) \rho \vec{V} \cdot d\vec{S} \right]_{post} = \left[\left(e + \frac{V^2}{2} + \frac{p}{\rho} \right) \rho \vec{V} \cdot d\vec{S} \right]_{pre}$$

But,

$$h = e + \frac{p}{\rho}$$

Hence the energy equation can be written as,

$$\left[\left(h + \frac{V^2}{2} \right) \right]_{post} = \left[\left(h + \frac{V^2}{2} \right) \right]_{pre}$$

or

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

$$h_1 + \frac{u_1^2 + w_1^2}{2} = h_2 + \frac{u_2^2 + w_2^2}{2}$$

But we know that, $(w)_{pre} = (w)_{post}$ or $w_1 = w_2$

Hence,

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$$

Energy equation is also similar as that of energy equation for normal shock. Here 'u' is the velocity normal to the shock. From mass, momentum and energy equations, it is clear that, only velocity normal to the shock, is responsible for change in all the properties. Hence we can still use all the equation of static and total property ratios derived for normal shock relations by changing the freestream Mach number to Mach number normal to the shock.

If freestream or upstream Mach number and the shock angle are known, then we can calculate the Mach number normal to the shock as,

$$M_n = M_1 \sin \beta$$

Prandtl's relation for the oblique shock is,

$$\frac{1}{M_{n1}^2} = M_{n2}^2$$

This relation suggests that, for oblique shock, normal Mach number before or upstream to the shock is supersonic and hence normal Mach number after or downstream to the shock is subsonic.

The static property ratios for oblique shock are,

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_n^2}{2+(\gamma-1)M_n^2}$$

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}(M_n^2 - 1)$$

$$\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma+1}(M_{n1}^2 - 1) \right] \left[\frac{2 + (\gamma-1)M_{n1}^2}{(\gamma+1)M_{n1}^2} \right]$$

Total property ratios can be re-written in the same way. We can as well calculate the Mach number behind the shock as,

$$M_{n2}^2 = \frac{1 + \left[\frac{(\gamma+1)}{2} \right] M_{n1}^2}{\gamma M_{n1}^2 - \frac{(\gamma-1)}{2}}$$

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)}$$

θ - β -M relation

It has been already observed that the Mach number normal to the shock is responsible for all the property variations for given shock angle. However this shock angle can be easily calculated from the upstream or freestream Mach number for given wedge or deflection angle. Consider the same control volume shown in **Fig.** Relation between velocities and angles before and after the shocks are,

$$\tan \beta = \frac{u_1}{w_1} \quad \text{Before the shock}$$

$$\tan(\beta - \theta) = \frac{u_2}{w_2} \quad \text{After the shock}$$

But we know that,

$$w_1 = w_2$$

Hence,

$$\frac{\tan(\beta - \theta)}{\tan \beta} = \frac{u_2}{u_1}$$

But

$$\rho_1 u_1 = \rho_2 u_2 \text{ hence, } \frac{u_2}{u_1} = \frac{\rho_1}{\rho_2}$$

Therefore,

$$\frac{\tan(\beta - \theta)}{\tan \beta} = \frac{\rho_1}{\rho_2}$$

From density ratio Eq (5.4), we have

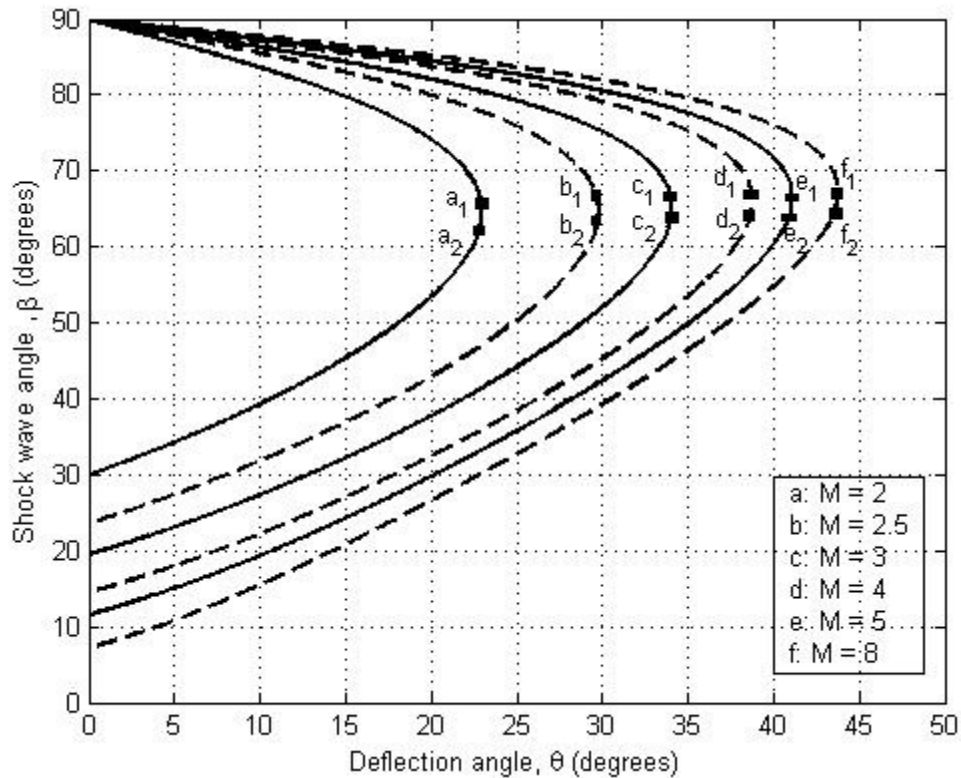
$$\frac{\tan(\beta - \theta)}{\tan \beta} = \frac{(\gamma + 1)M_{n1}^2}{2 + (\gamma - 1)M_{n1}^2}$$

From the expression of upstream Mach number, $M_n = M_1 \sin \beta$,

$$\frac{\tan(\beta - \theta)}{\tan \beta} = \frac{2 + (\gamma - 1)M_1^2 \sin^2 \beta}{(\gamma + 1)M_1^2 \sin^2 \beta}$$

$$\tan \theta = 2 \cot \beta \left[\frac{M_1^2 \sin^2 \beta - 1}{M_1^2 (\gamma + \cos 2\beta) + 2} \right]$$

This is the expression between upstream Mach number, shock angle and wedge angle. In most general case, we need to know the shock angle for given Mach number and wedge angle. Following figure provides the information about the same. In this figure, each curve corresponds to various possible shock angles are for an given Mach number and flow deflection angle.

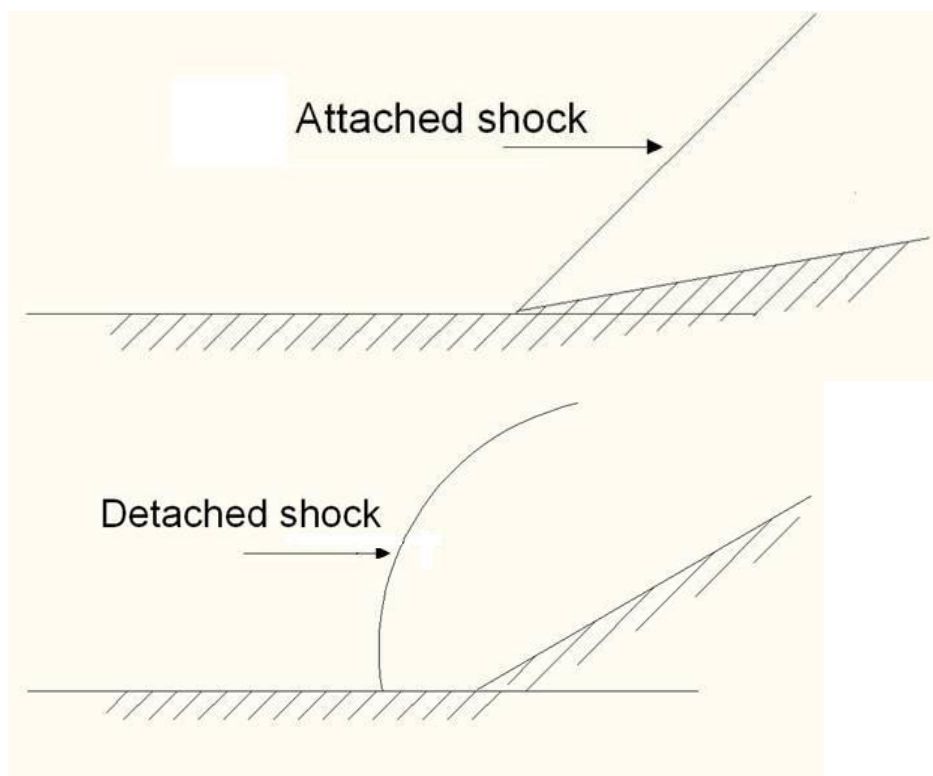


θ - β - M relation

Following points should be noted from this figure.

1. For a given Mach number and wedge angle, there are two possible shock angles. The lower shock angle corresponds to a weak shock solution and higher shock angle corresponds to strong shock solution. Weak shock solution is the most familiar solution or situation we encounter in the nature. Higher shock angle values, obtained out of two possible solutions for the given Mach number and deflection angle, is referred as the strong shock solution. The strong shock solution may exist for a given Mach number and deflection angle if the pressure on the wedge can be increased independently. However this situation is most uncommon. Points having subscript 1 in **Fig.** are the reference points for the known Mach number above which shock is said to be strong shock and below which shock is said to be weak shock. Points having subscript 2 in **Fig.** are the reference points for the known Mach number above which shock flow behind the shock is subsonic and below which flow behind the shock is supersonic.

2. If we consider the weak shock solution, then for a given Mach number, increase in deflection angle increases the shock angle. Thus Increased shock angle decreases the Mach number behind the shock. Hence shock strength increases for a given Mach number with increase in deflection angle. For a particular deflection angle, we get, maximum shock angle for a given Mach number above which there is no attached shock solution. Hence there exists no solution or deflection angle for that Mach number for which shock is attached. This situation is discribed in **Fig.**



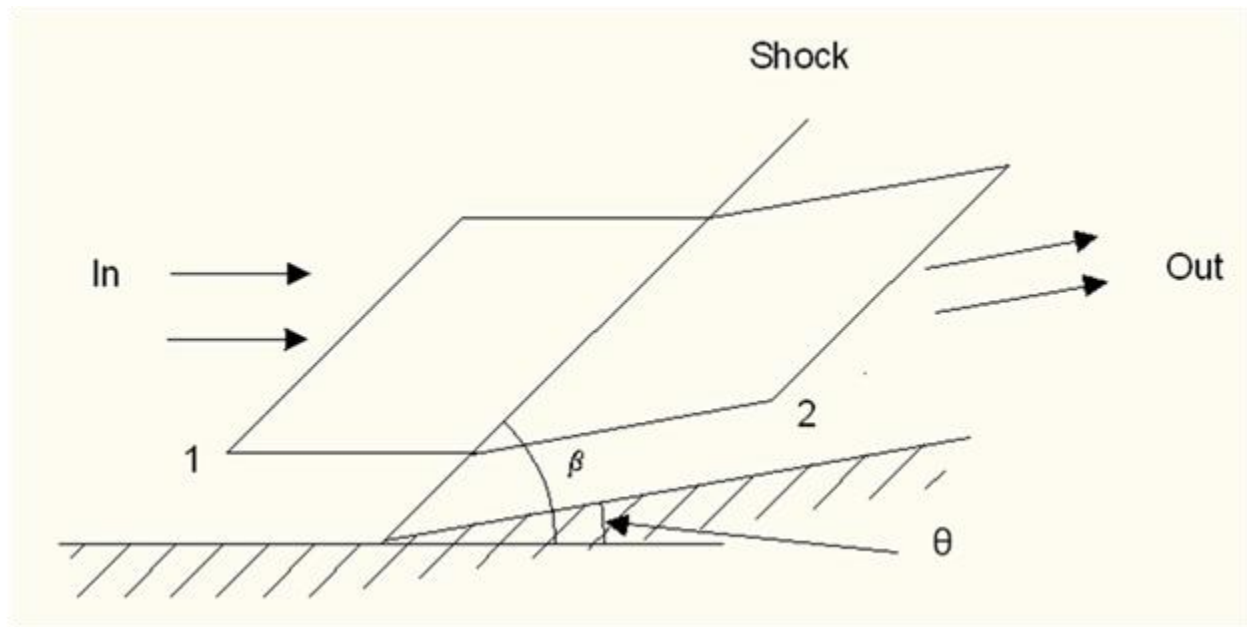
Attached and detached shock solutions.

3. If we consider the weak shock solution, then, for a given deflection angle, increase in Mach number decreases the shock angle and shock becomes weaker.
4. All the curves for various Mach numbers can be seen to meet at 90 degree shock angle for zero degree deflection angle. This situation corresponds to normal shock. However, the other solution for zero degree deflection angle corresponds to Mach angle, $\mu = \sin^{-1}(1/M)$.

Therefore intersection of the curve with shock angle for weak solution for all the Mach number is different.

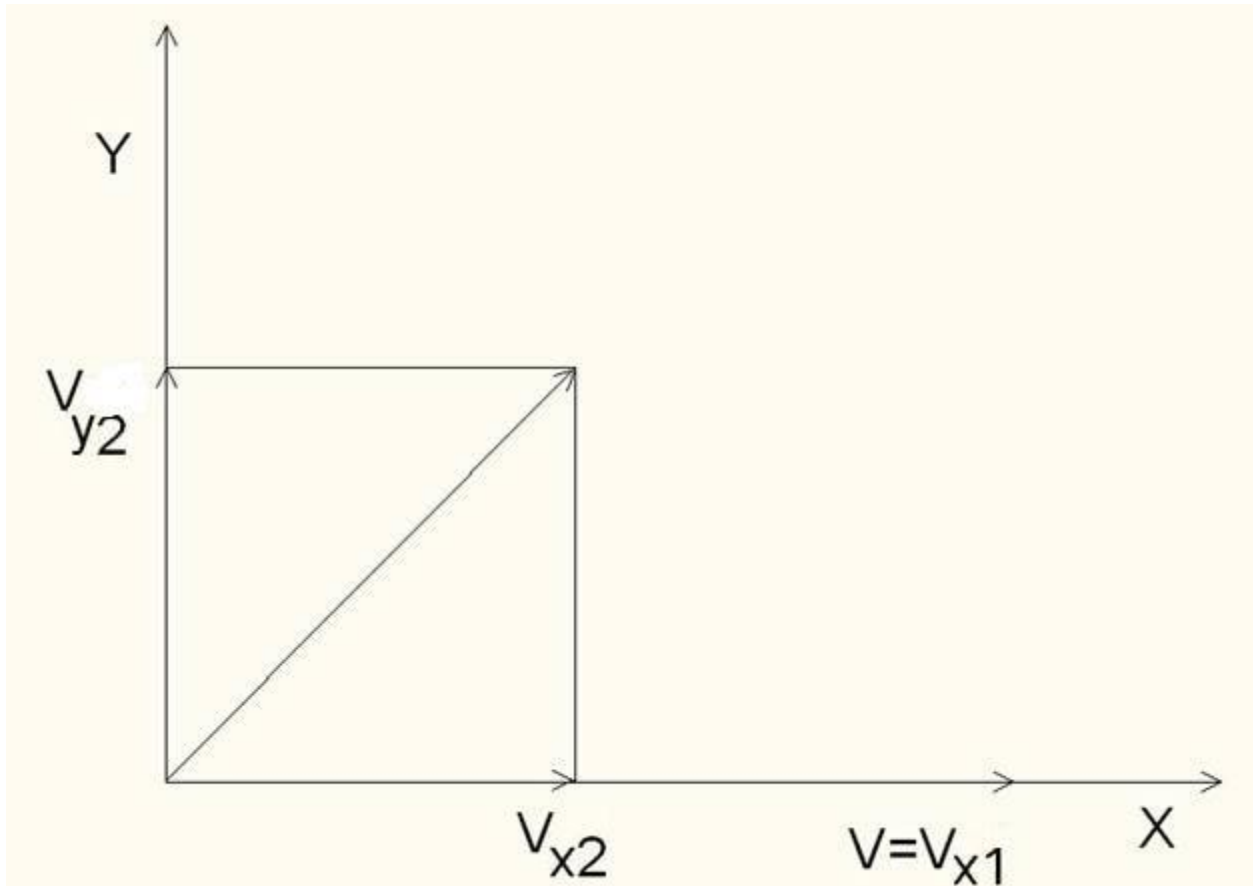
Shock Polar

In this section we are going to discuss about the graphical representation of pre and post shock velocities. Consider the supersonic flow of Mach number M_1 passing over a wedge of deflection angle θ as shown in **Fig.**



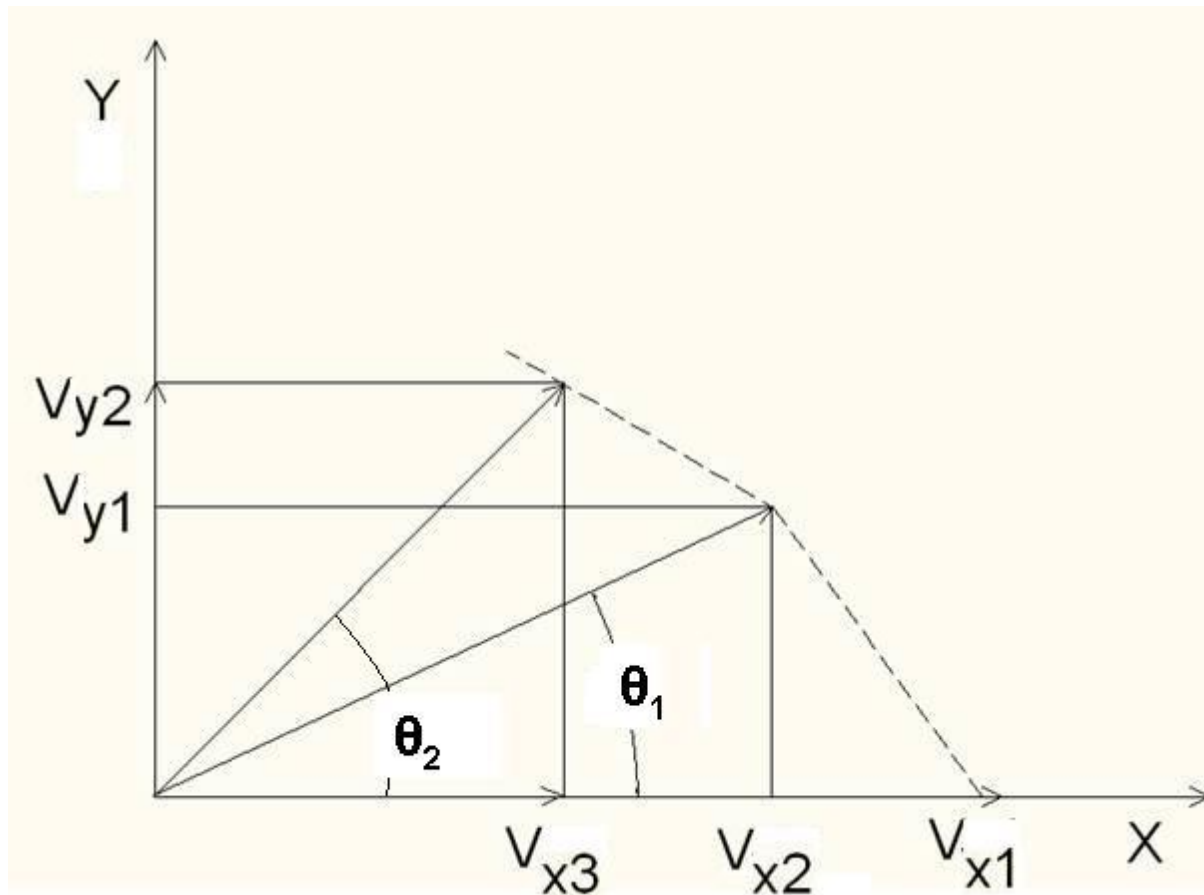
Supersonic flow over a wedge

Let V_x be the component of velocity in x-direction while V_y be the component of velocity in y direction. Hence V_{x1} is the x-direction velocity component for upstream velocity and V_{x2} is the x-direction velocity component for post shock velocity. Similarly V_{y1} and V_{y2} are the y-direction velocity component for upstream and post shock velocities respectively. If the upstream velocity is parallel to x axis then V_{y1} is equal to zero since V_{x1} is equal to V_1 . Now let's plot velocities along their respective directions as shown in **Fig.** Here the resultant velocity V_1 is parallel to x axis while V_2 makes an angle θ with x-axis, which is the flow deflection angle.



Velocity plot for shocked flow

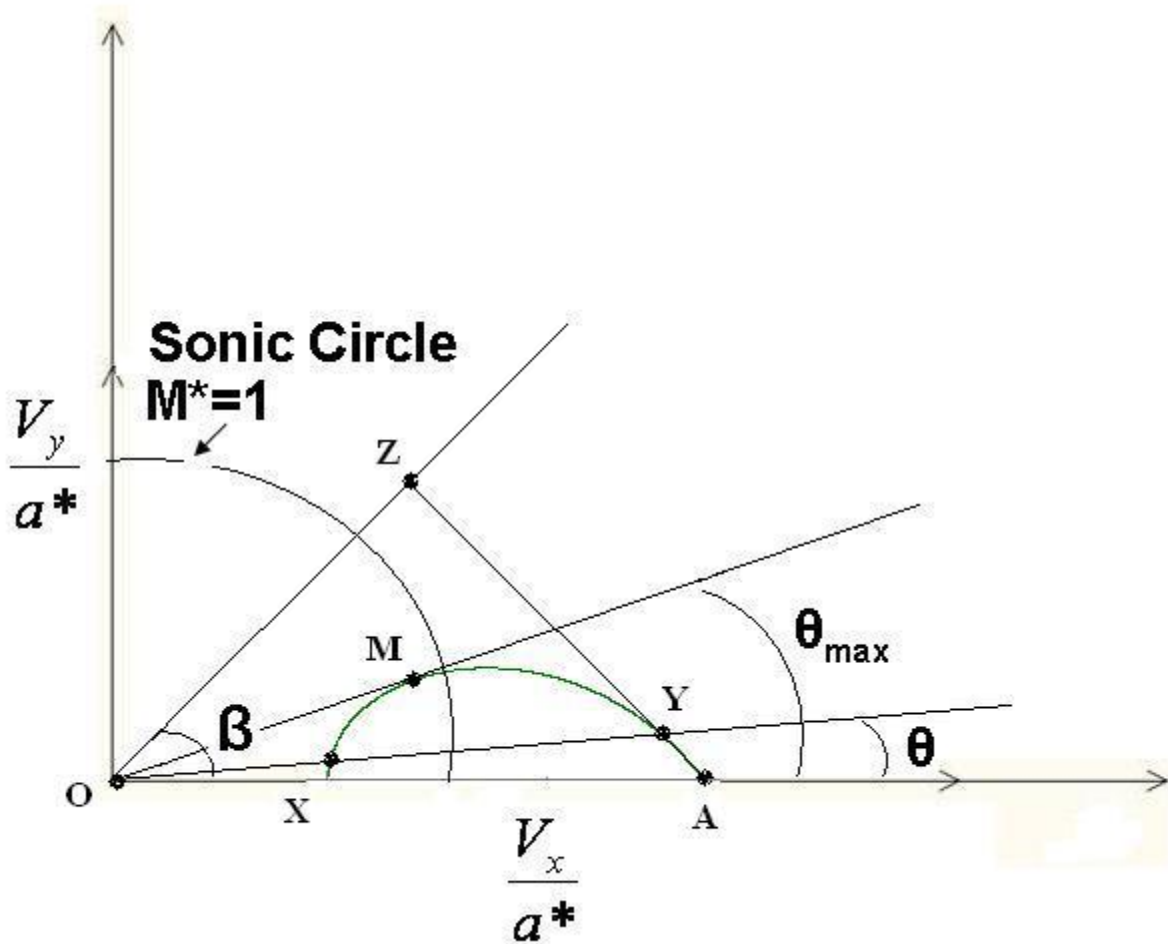
Now if we increase the deflection angle then V_{x2} will decrease and V_{y2} will increase since in such cases resultant velocity V_2 will have to make an increased angle θ with the x-axis. Figure represents the same process. In the same way if we change the wedge angle for all possible attached shock solutions and join them together then such a plot is called as hodograph.



Velocity plot for various shocked flows

In the same way we can plot for various freestream Mach number for all the possible deflection angles. However, if we increase the freestream Mach number by increasing freestream velocity then it becomes impossible to plot such a graph when V_{x1} becomes ∞ for M_1 equal to ∞ . To make such a plot possible, let's divide x axis and y axis by a^* which is the reference or starred quantity. This non-dimensionalisation makes it possible to represent V_{x1} since it will be represented by M_1^* . We can divide by a^* to post shock velocities as well, since a^* is constant in the flowfield for an adiabatic flow. Hence V_2 gets transformed to M_2^* which will make an angle equal to the flow deflection angle with the x-axis. This plot is called as shock polar. Such a shock polar can be plotted for very high Mach numbers M_1 also due to the fact that, if M_1 is equal to ∞ , M_1^* is equal to 2.54. Therefore this plot is equally helpful for representing the pre and post shock

velocities, flow deflection angle and the shock angle. Typical shock polar is shown in **Fig.**



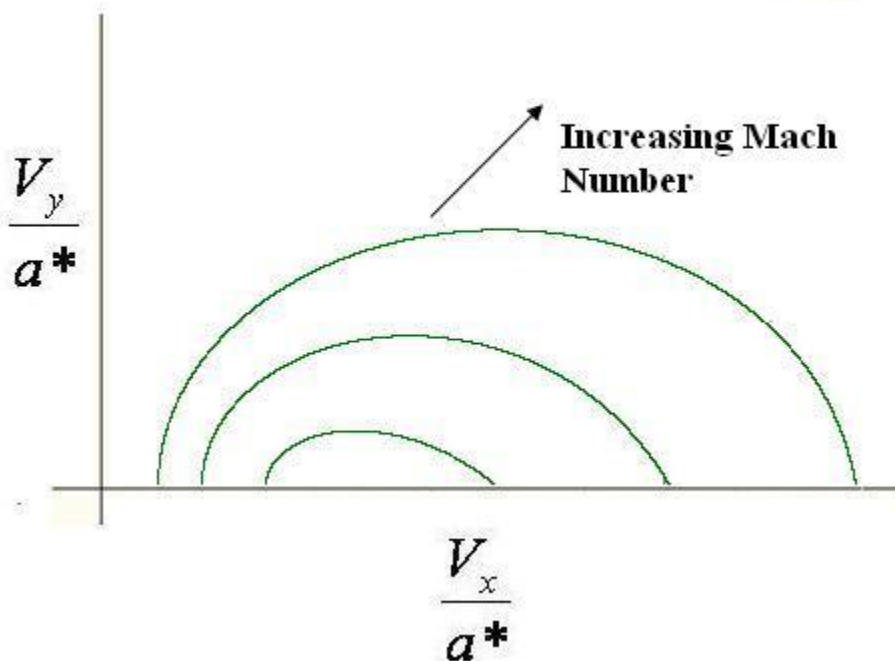
Typical Shock Polar

Analysis of Shock Polar

Shock polar can be used as a tool to evaluate the post shock properties from know pre shock conditions. We can calculate a^* and hence M_1^* which is the required pre shock input condition from the known freestream conditions,. Hence using flow deflection angle θ and M_1^* , we can evaluate the post shock condition as, M_2^* using a given shock polar. Consider a shock polar given in **Fig.**Any line,

say OXY, drawn from origin which intersects the given shock polar at two locations, X and Y. Point Y of line OXY represents higher value of M_2^* than the point X. Hence point Y represents the weak shock solution while point X on shock polar represents the strong shock solution. Angle made by line OXY gives the flow deflection angle. To find out the shock angle, we have to draw a line AY and extend the same till it intersects a line (OZ) drawn from the origin at 90 degree. The angle made by line OZ with x-axis is the shock angle since this arrangement satisfies the constraint of equality of component of velocity parallel to the shock before and after the shock. If we draw any line OM, tangent to the shock polar, then the angle made by the line OM with x-axis represents maximum deflection angle possible for the given freestream Mach number. If we draw a circle of radius equal to a^* then such a circle cuts the shock polar at N which represents the Mach number equal 1 behind the shock.

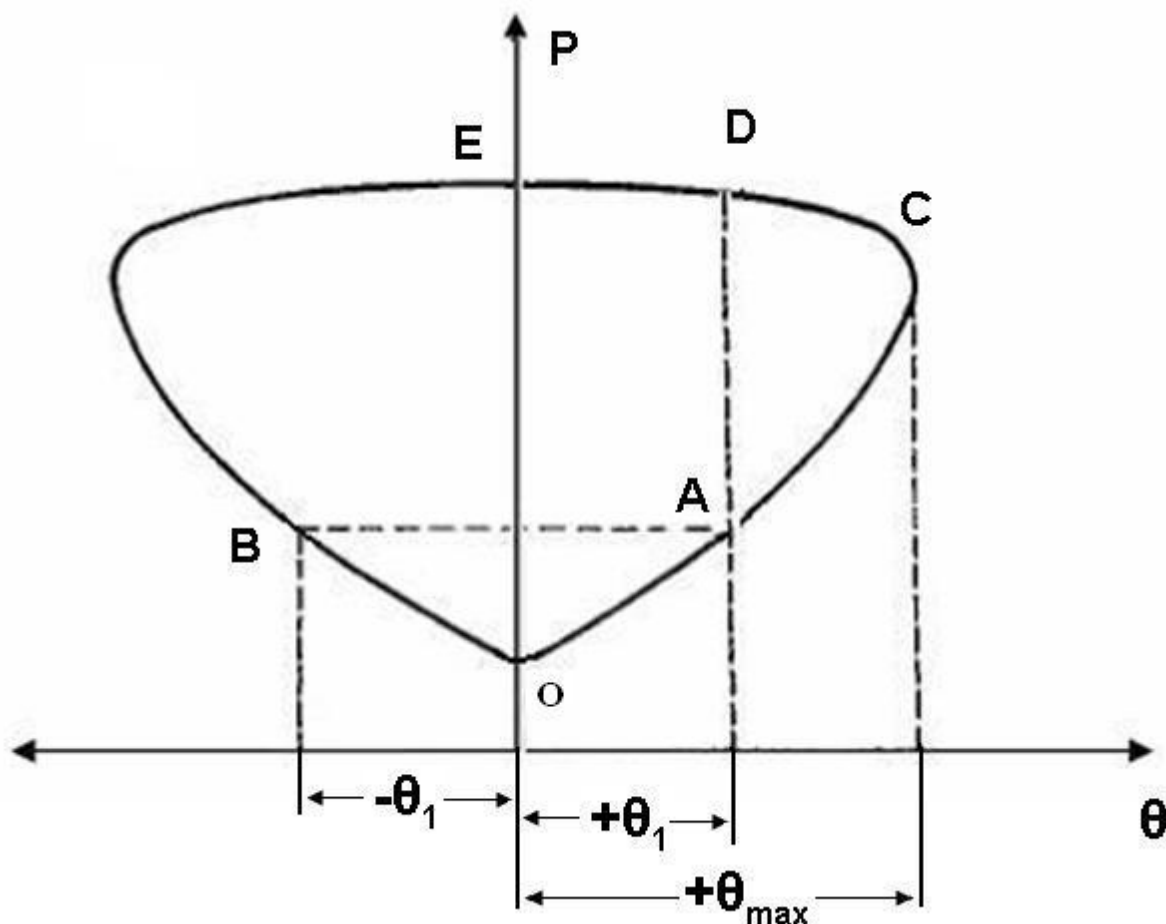
Shock polar for various Mach numbers is shown in **Fig.**



Shock polar for various Mach numbers

P- θ diagram

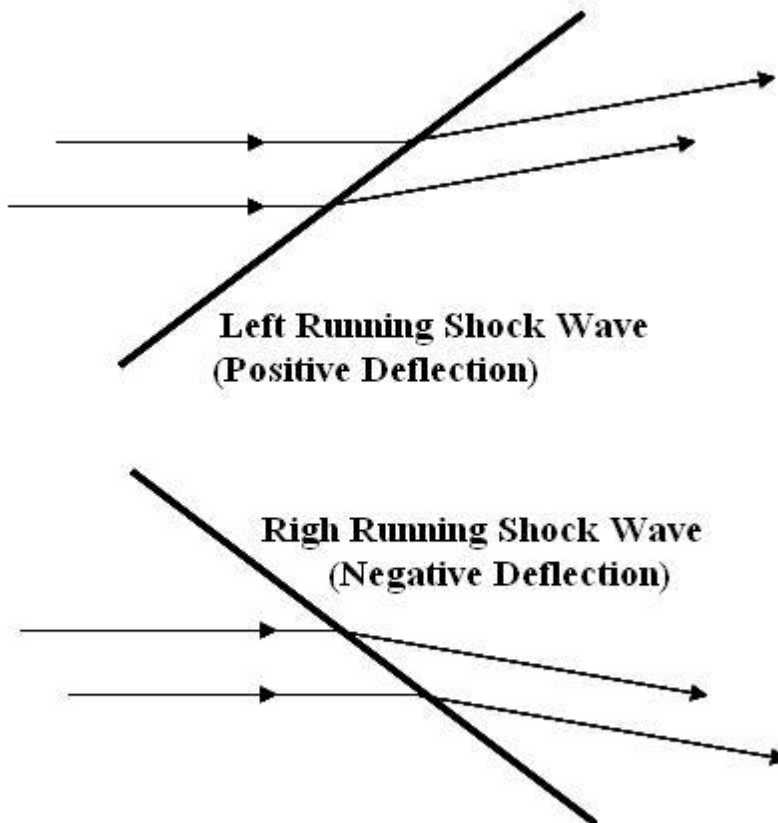
Pressure-deflection diagram is one more tool to present the oblique shock solution. This diagram necessarily plots various possible pressures for change in flow deflection angle for a given Mach number. Hence we can plot this, P- θ diagram, for all the Mach numbers to evaluate the range of pressure for a given Mach number and the same for given deflection angle. This diagram also provides the shock detachment angle. Typical P- θ diagram for a known Mach number is shown in **Fig.** All the points marked on this diagram have some special feature associated with them.



Typical Pressure deflection diagram for a known Mach number

Here point O represents the pre-shock conditions and corresponds to the conditions at zero degree flow deflection. The main feature of the feature is the

terminology used while constructing it. Here deflection of the flow by left running shock is treated as positive while deflection of the flow by right running shock wave is considered as negative. The typical left and right running shock waves are shown in **Fig.**



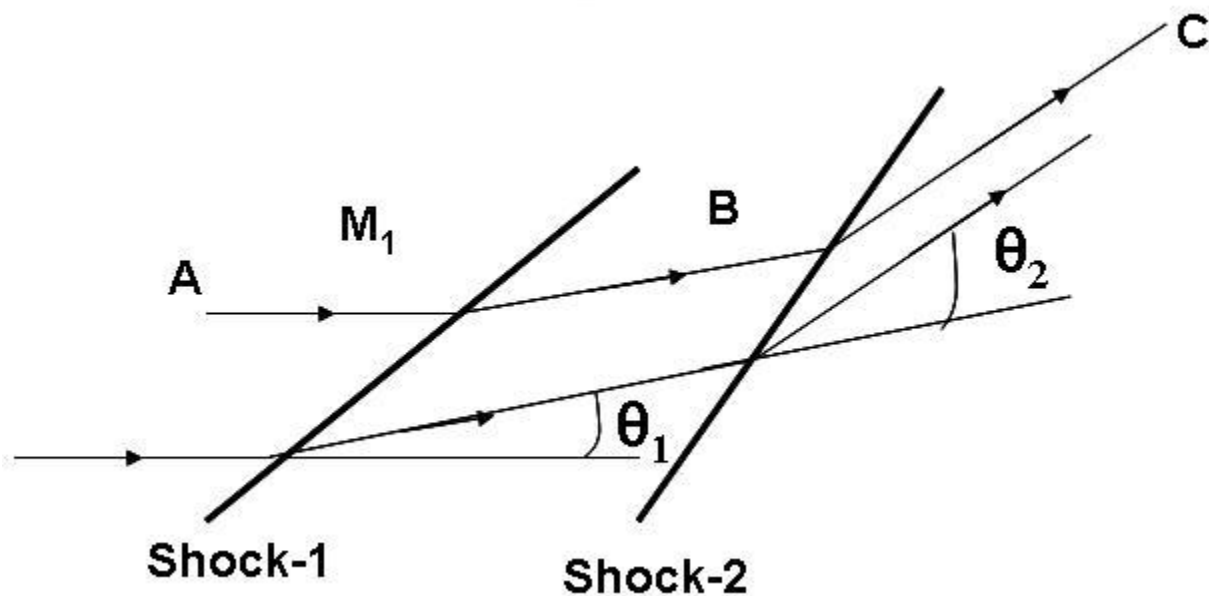
Typical left and right running shock waves

The shock wave is termed as a left running shock wave where we can feel the shock wave to be passing on left side of the observer who is standing on the shock wave with downstream attention. Similarly shock wave is termed as a right running shock wave where we can feel the shock wave to be passing on right side of the observer who is standing on the shock wave with downstream attention. Therefore, the points A and B of the **Fig.** represent same post shock pressure for the same pre shock conditions but for two different deflections viz. positive θ and negative θ respectively. Typical shock patterns represented by these two points can be visualised using **Fig**. Point C on this P- θ curve

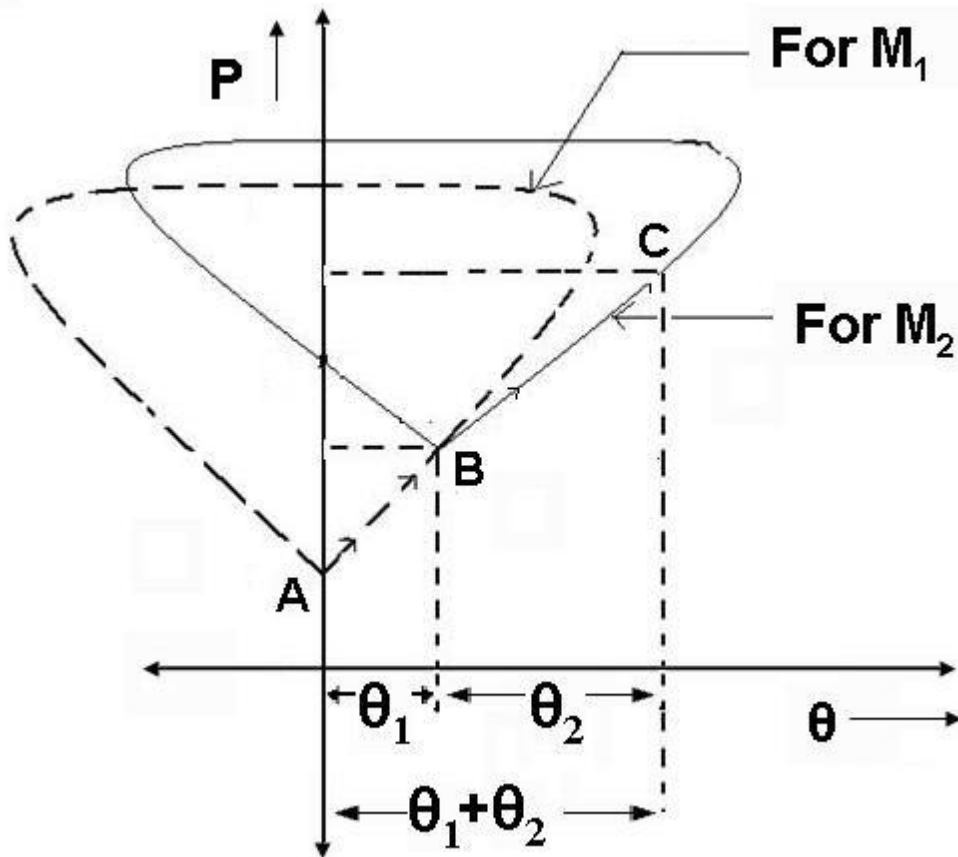
represents the maximum possible θ or flow deflection angle. Point D represents higher pressure for the same θ for a given Mach number as that represented by A. Therefore, it becomes clear to us that point A is the weak shock solution while point D is the strong shock solution since pressure corresponding to point D is more than that of A. In the same diagram, point E represents zero deflection and highest pressure which corresponds to the normal shock situation which leads to no deflection with maximum pressure rise for the given Mach number.

Applications of P- θ diagram

Pressure-deflection diagram is very useful for understanding various flow features like shock-shock interaction, shock reflection etc. Therefore the main objective of working with this diagram is to understand the deflection of flow which encounters various shocks. Consider a supersonic flow experiencing two left running shock waves shown in **Fig.** Corresponding P- θ is shown in **Fig.**



Supersonic flow passing through two left running shock waves

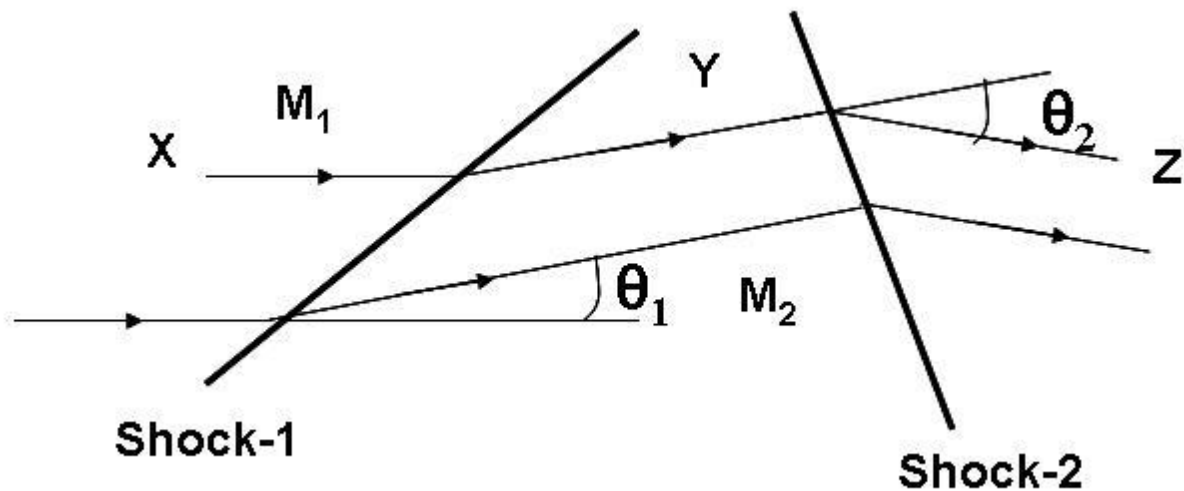


P-θ diagram for the supersonic flow passing through two left running shock waves

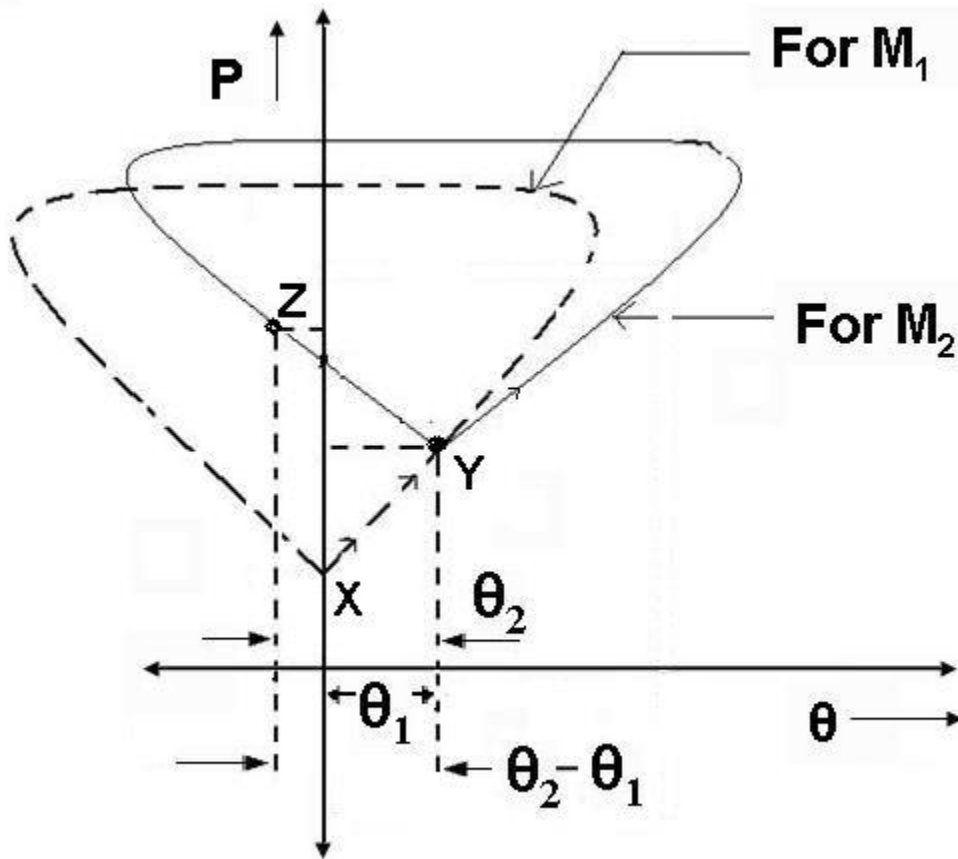
Here we can clearly understand the usefulness of this diagram. Consider a streamline ABC in this flowfield. The initial or pre shock conditions are given point A on P-axis which corresponds to zero deflection. After passing through the first shock, the flow gets deflected by an angle θ_1 which is represented by point B. We have to follow the **P-θ** diagram starting from point A, corresponding to its freestream conditions, to get the point B. The point B clearly portrays increase in pressure and positive deflection of the flow in the presence of left running shock wave. Further experience of next shock brings in the change in flow properties which are represented by point C. We have to draw a new **P-θ** diagram at point B, corresponding to its freestream conditions, to arrive at point C. For this fact, deflection of the flow at point B would be treated as zero and relative displacement between point B and C (θ_2) should be plotted to get the point C.

However state of the point C given in this diagram gives total flow deflection and final pressure.

Consider a supersonic flow experiencing a left running shock wave followed by a right running shock wave shown in **Fig.** Corresponding P- θ is shown in **Fig.** Consider a streamline XYZ in this flowfield.



Supersonic flow passing through through left and right running shock waves respectively



P- θ diagram for the supersonic flow passing through left and right running shock waves respectively

The initial or pre shock conditions are given point X on P -axis which corresponds to zero deflection. After passing through the first shock which is left running shock, the flow gets deflected by an angle θ_1 which is represented by point Y . Here as well, we have to follow the **P- θ** diagram starting from point A , corresponding to its freestream conditions, to get the point Y . The point Y clearly portrays increase in pressure and positive deflection of the flow in the presence of left running shock wave. Further experience of right running shock wave brings in the change in flow properties which are represented by point Z . We have to draw a new **P- θ** diagram at point Y , corresponding to its freestream conditions, to arrive at point Z . For this fact, deflection of the flow at point Y would be treated as zero and relative displacement between point Y and Z (θ_2) should be plotted to get the point Z . Since the flow is facing right running shock

wave, the deflection θ_2 is negative in reference with the origin at Y. However presence of the shock increases pressure of the flow. Therefore, state of the point Z given in this diagram gives total flow deflection and final pressure.

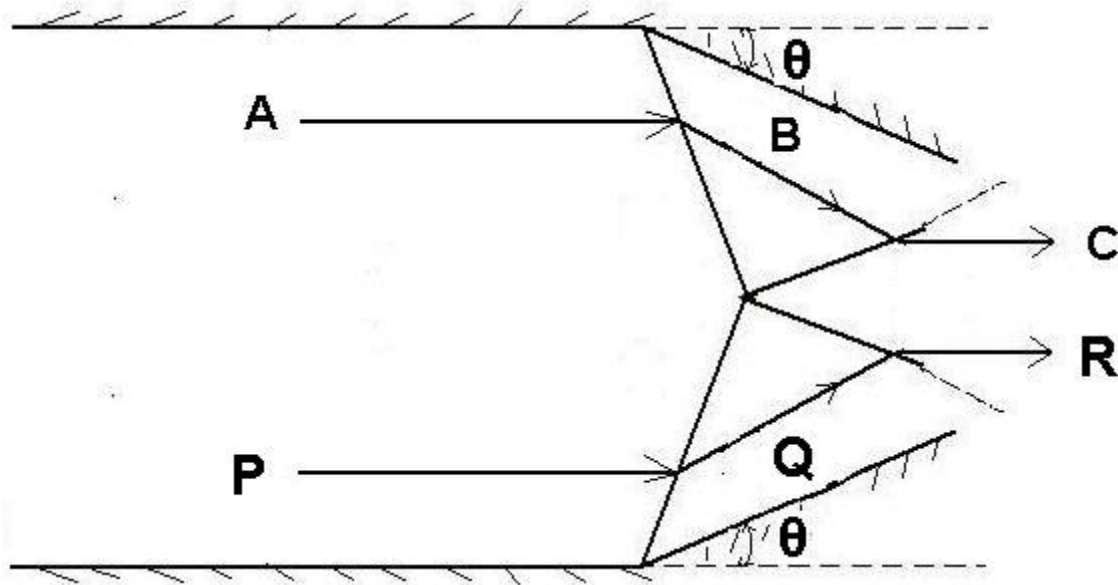
Shock-Shock Interaction

Interaction of two shocks is of great interest for various reasons. The major reason is the prediction of the flow field after interaction. This flowfield is greatly dependant on the the family of the shock (right running or left running) and strength of the shocks in turn freestream properties. Aerodynamics of the body highly dependant on the shock shock (S/S) interaction. Edney* has defined six types of shock shock interactions based on family and strength of the interacting shocks and post shock flowfield. Herewith, basics of the shock shock interaction are mentioned.

* B. Edney, Anomalous heat transfer and pressure distributions on blunt bodies at hypersonic speeds in the presence of an impinging shock, Aeronautical Research Institute of Sweden, Report 115, Stockholm (1968).

Interaction of shocks of different family

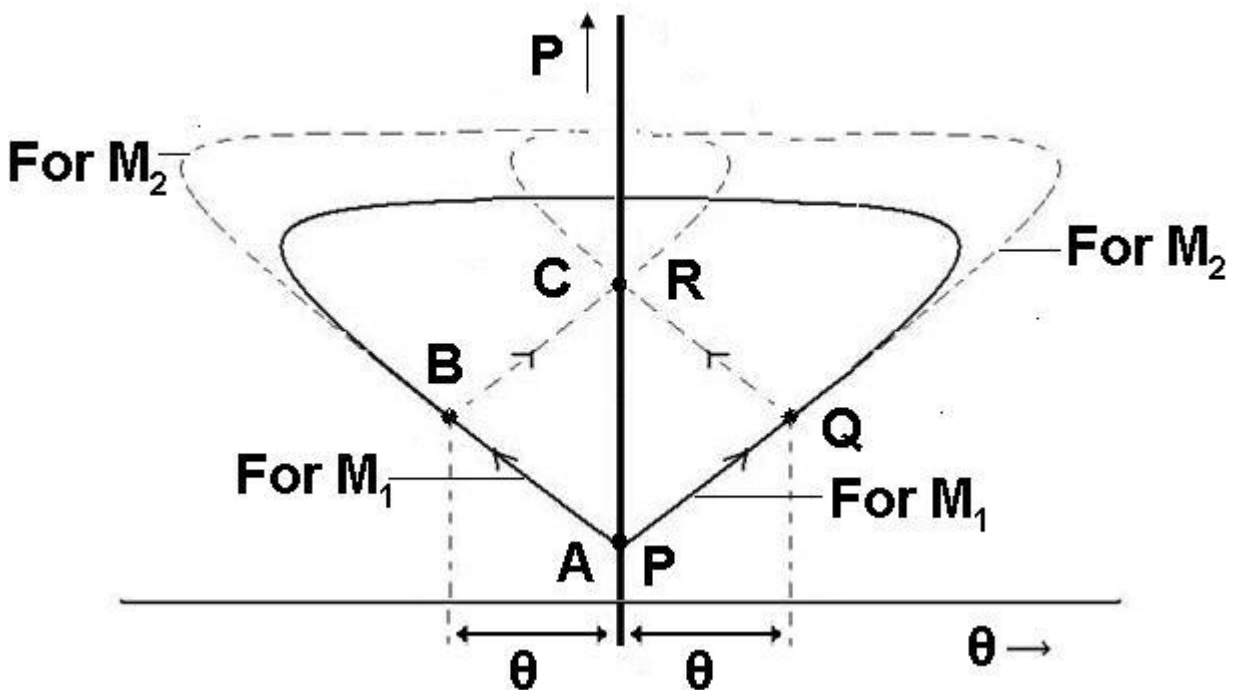
Lets consider the situation given in **Fig.** to understand the S/S interaction for shocks from two different families.



Typical example of S/S interaction of shocks of different families and same strength.

Here the right running shock originated from the top wedge interacts with the left running shock originated from the bottom wedge, hence this is the S/S interaction for shocks of different families. For simplicity we have considered flow deflection angles to be same (θ) for both the wedges. The top wedge, in this situation, is responsible for right running shock wave while bottom wedge is responsible for left running shock wave. Now consider the streamline ABC passing through the right running shock and PQR passing through the left running shock. The right running shock wave induces negative deflection to the streamline ABC at B, while left running shock wave induces positive deflection to the streamline PQR at Q. The corresponding increase in pressure associated with the flow turning are shown in **Fig.** using the P- θ diagram. Here we can clearly see that points A & P are identical in this plot, since both the points are in the region upstream of shocks. Since streamline ABC gets negative deflection and streamline PQR gets positive deflection of same magnitude, points B and Q represent same pressure but different deflections of same magnitude. Suppose, both the shocks are weak enough to keep the flow supersonic in the region given by points B and Q of the streamlines. The flow properties will be identical

downstream of the first shocks (at B and Q), except the y-direction velocity. After interaction we can see that two different shocks originate from the interaction point. The main reason for this shock formation is to avoid the collision and intersection of streamlines ABC and PQR. Therefore, streamline ABC passes through a left running shock and streamline PQR passes through the right running shock wave. Hence the further pressure rise is evident in **Fig.** in the presence of post interaction shocks.

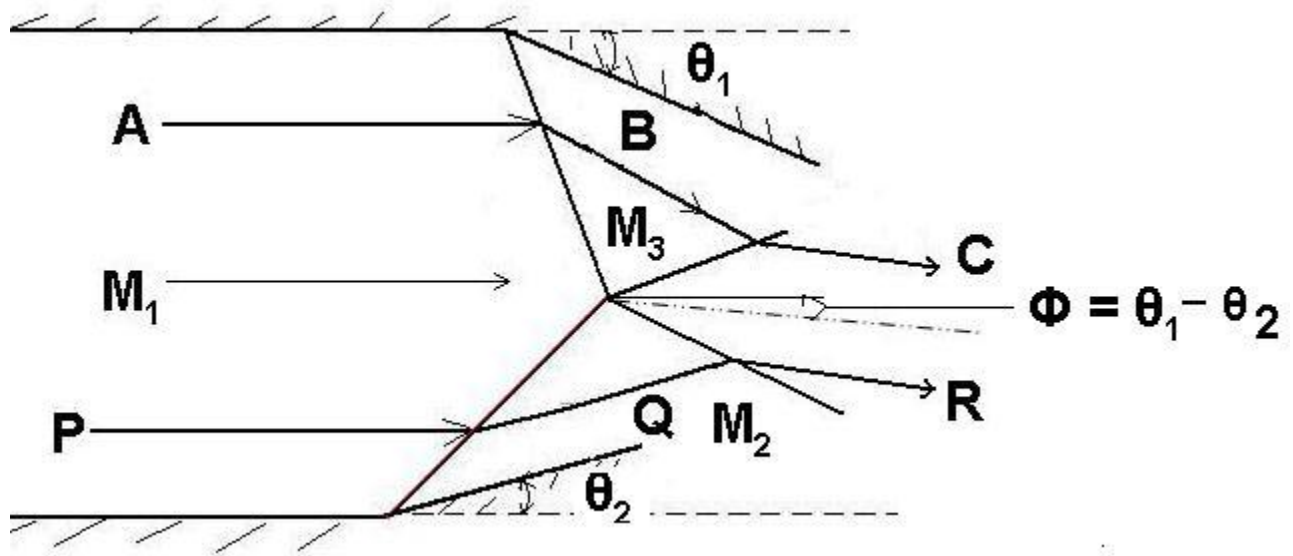


Pressure deflection diagram for S/S interaction of shocks of different families and same strength.

Moreover, these shocks cancel out the deflection so as to make the both the streamlines parallel to each other without avoiding their crossing. For flow downstream of first shock, we have to draw the pressure-deflection diagram from points B & Q respectively corresponding to their Mach numbers (which are identical in the present situation). Again it can be seen that, points C and R represent same pressure and deflection.

Interaction of shocks of different family

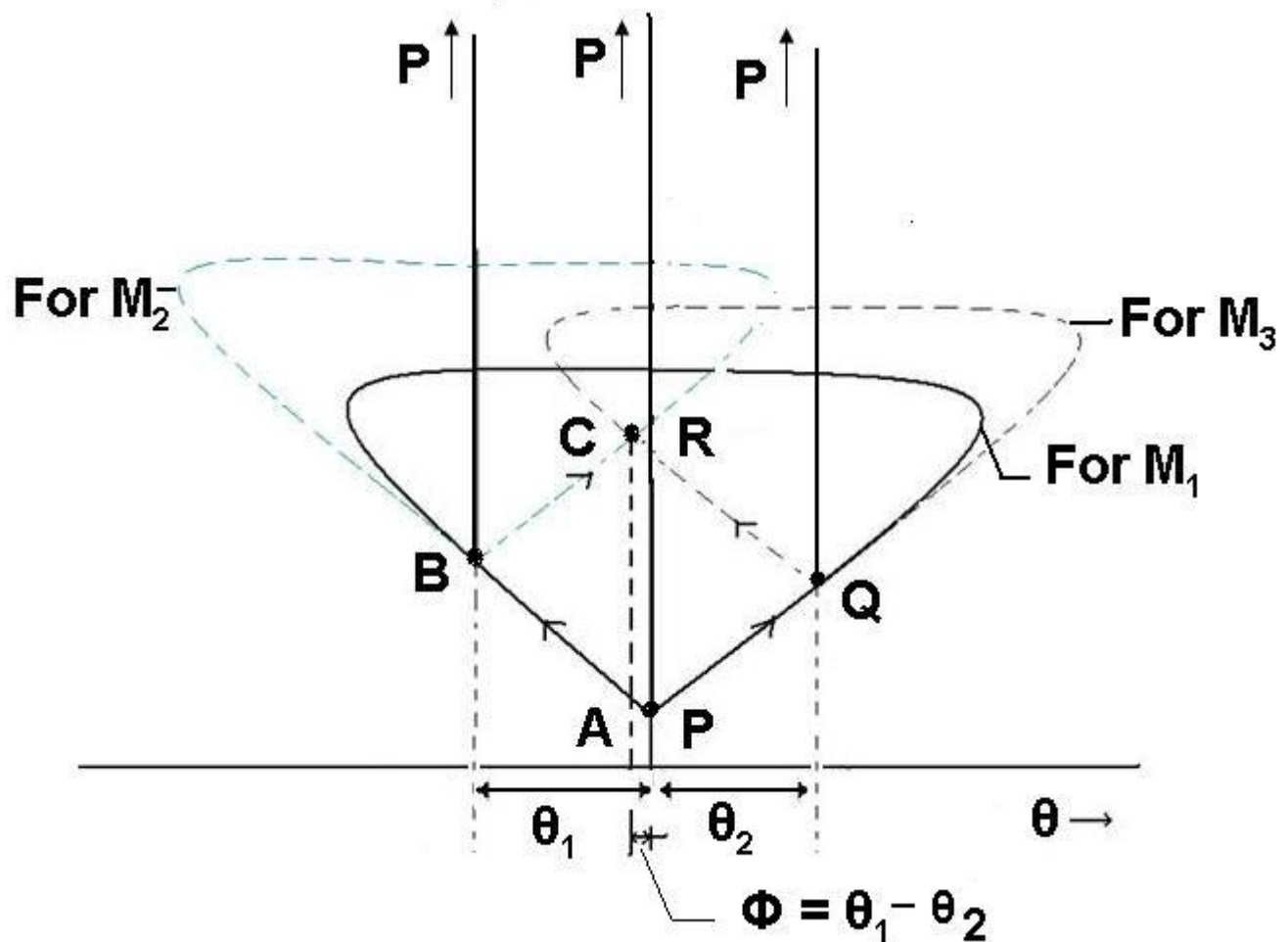
Now consider the same S/S interaction problem for shocks of different strengths. As shown in **Fig** wedge angles and hence the flow deflection angles are shown different to imitate the two different shock strength for S/S interaction understanding.,/



Typical example of S/S interaction of shocks of different families and different strength.

Here the wedge angles are θ_1 and θ_2 for top wedge and bottom wedge respectively, where $\theta_1 > \theta_2$ in magnitude. Similar to the earlier situation, the top wedge is responsible for right running shock wave while bottom wedge is responsible for left running shock wave. Consider the streamlines ABC passing through the right running shock originated from top wedge and streamline PQR passing through the left running shock originated from bottom wedge. Initial deflection acquired by both the streamlines will be different in this case since the flow deflection angles are different. Flow deflection angles and corresponding pressures at point B and Q are shown using the P- θ diagram in **Fig**. Here we can

clearly see that points A & P are identical due to the fact that both the points belong to the region upstream of shocks.



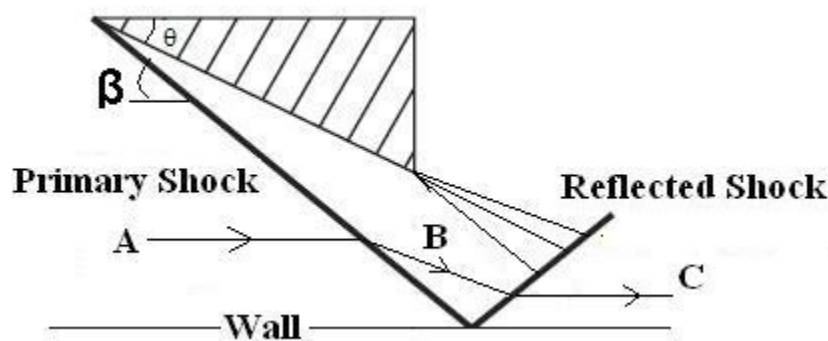
Pressure deflection diagram for S/S interaction of shocks of different families and different strength.

In this situation, streamline ABC gets negative deflection of higher magnitude at points B in comparison with the deflection of streamline PQR at point Q. Due to higher strength of the right running shock, pressure at point B will be more than that of point Q. Therefore unlike earlier situation, all the flow properties will be different downstream of the first shocks at B and Q. Suppose, both the shocks are weak enough to keep the flow supersonic in the region given by points B and Q of the streamlines. However, when both (right and left running) shocks intersect, we can see similar interaction from where two new shocks originate.

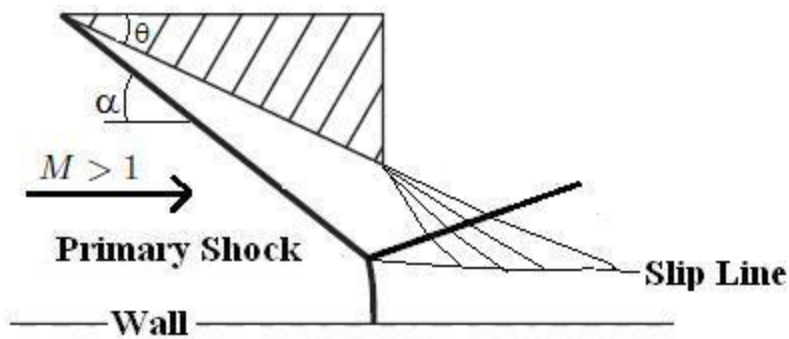
The reason for presence of these two shocks is to make the flows (passing over top wedge and passing over bottom wedge) parallel to each other. Streamline ABC which has higher deflection, now passes through the left running shock but can not cancel the earlier deflection completely. Hence, the part of the flow passed from the top wedge retains certain negative deflection in the region of point C. At the same time, the part of the flow passing from the bottom wedge, which has positive deflection with respect to the free stream, passes through the right running shock which then cancels its positive deflection and induces negative deflection in the region of point R so that streamlines ABC and PQR become parallel to each other at the end of the interaction region. Hence the strength of the shock encountered by the streamline PQR while passing from Q to R is higher than the strength of the shock encountered by the streamline ABC while passing from B to C. The new entity which we can observe is the slip line (shown by dotted line) which originates from the point of intersection of the shocks. The main purpose of this line is to avoid mixing of two flows passing from two different shock patterns which have different entropies. However pressure and direction of velocity remains conserved across this line. Therefore the final points C and R shown in **Fig.** portray finite amount of same deflection and pressure.

Shock Reflection

Reflection of shock wave from the wall is mainly of two types, regular reflection and irregular or Mach reflection. These reflection patterns are shown in **Fig.**

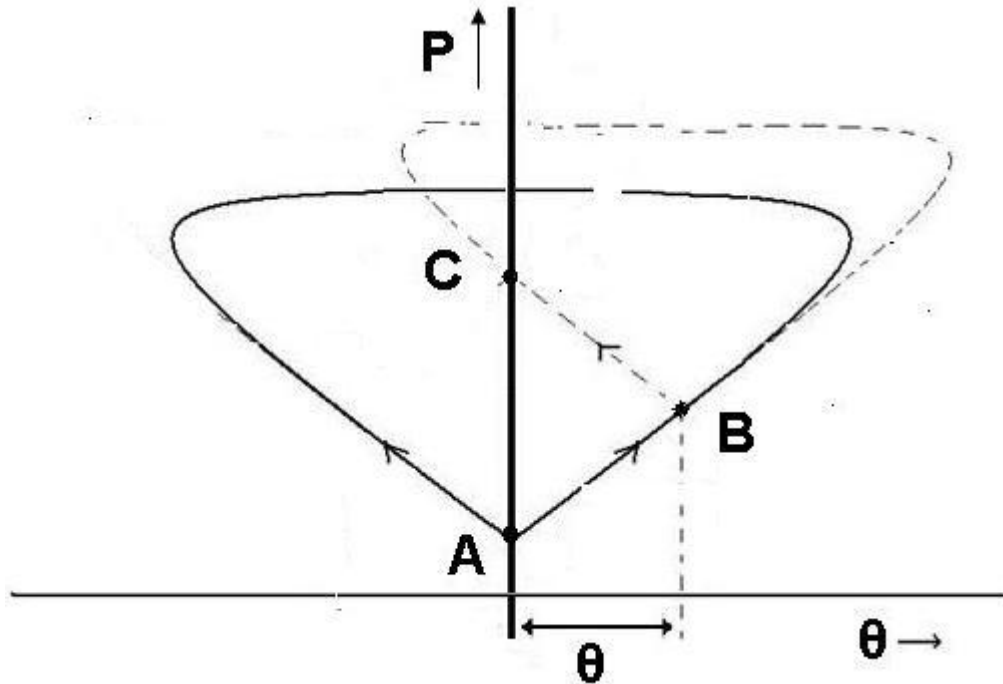


Typical regular reflection of shock wave



Typical Mach reflection of shock wave

The shock that originates from the corner of the compression wedge makes an angle β with the freestream velocity vector. Flow behind this shock gets deflected by angle θ and becomes parallel to the wedge. However, when the shock hits the top wall, the part of the flow which was parallel to the top wall, upstream of the shock, also gets deflected by angle θ . Hence the flow downstream of the shock does not remain parallel to the top wall. This contradicts the necessary condition of formation of shock, hence the primary shock gets reflected from the top wall to cancel out the deflection of the flow and to make it parallel to the wall. In this process shock reflection can be observed as the reflection of light ray, however the incident and reflection angle of shock are dependant on shock upstream Mach number. Therefore this reflection is called as the regular reflection. Typical pressure deflection diagram for the streamline ABC is shown in **Fig.**



Pressure deflection diagram for the regular reflection case

Providing equal and opposite deflection to the flow as that provided by incident shock is the objective of reflected shock. However possibility of such a deflection behind the reflected shock depends on the Mach number behind the incident shock or Mach number upstream to the reflected shock (in region B of **Fig.**). If the deflection required is more than the maximum possible deflection for that Mach number then the reflection of shock does not remain as regular reflection since the reflected shock gets detached from the wall.

Mach Reflection

Typical Mach reflection situation is shown in **Fig.** Lets understand the reason for deviation from regular reflection to Mach reflection. We know from **Fig.** that the presence of reflected shock is mainly to deflect the flow in region B and make it parallel with the wall. The amount of this flow deflection expected from the reflected shock is same as that from the primary or incident shock. However,

regular reflection ensures that this deflection is less than the maximum possible deflection corresponding to the Mach number in region B in the same figure. A situation has been shown in **Fig.** which shows the possibility of where the expected deflection using reflected shock more than the maximum deflection possible corresponding to Mach number in region B.

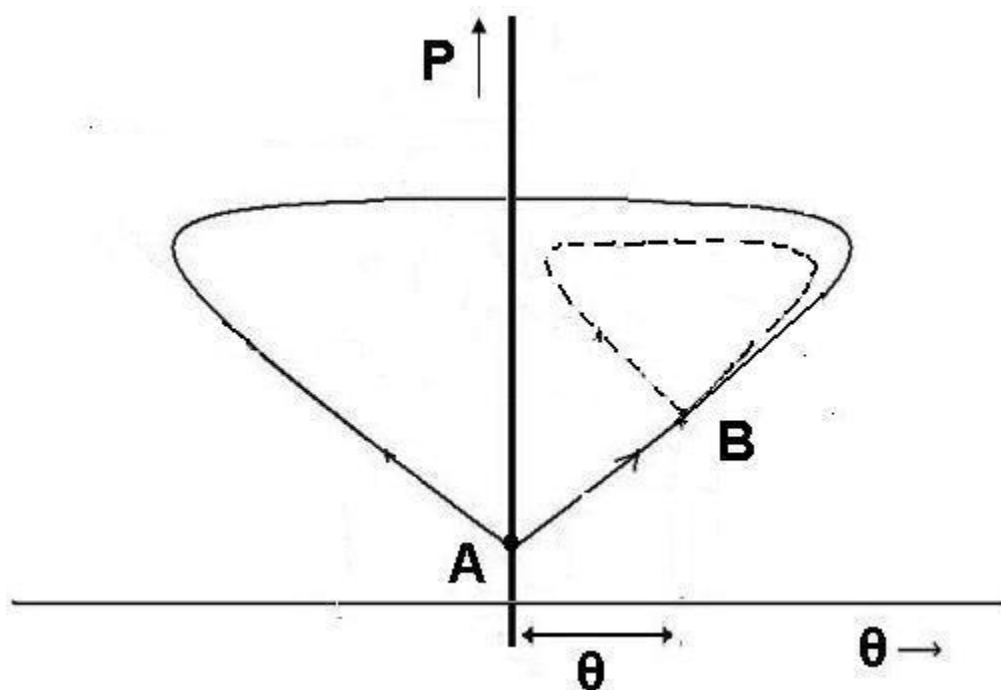


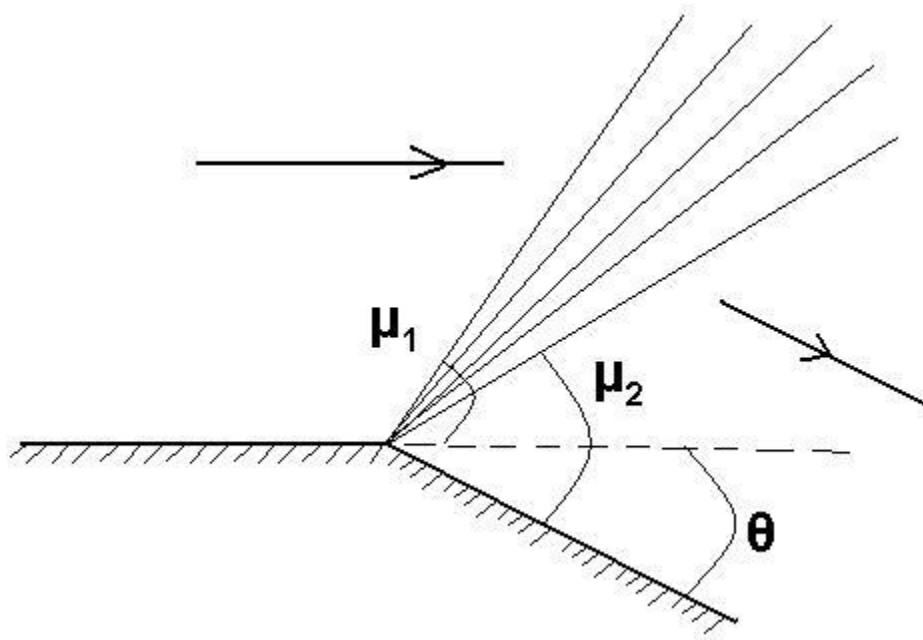
Fig. Reason for Mach reflection using pressure deflection diagram.

Therefore, since reflected shock can not provide the necessary deflection to the flow so as to make it parallel to the wall, the incident or primary shock does not reflect like the ray of light or regular reflection. This is the main reason for the Mach reflection shown in **Fig**

Expansion Fan

We have already seen that compression of supersonic flow takes place while passing through the shock. In other words, when the supersonic flow turns into itself then it undergoes the compression through a shock. Exactly opposite situation can be

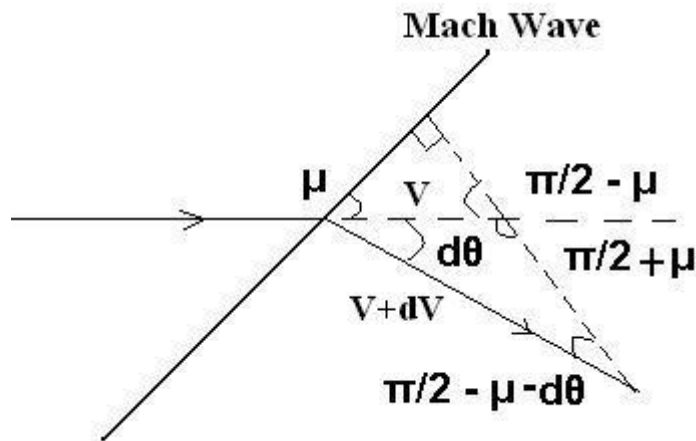
encountered when the supersonic flow turns out of itself where, expansion of the supersonic flow takes place. This expansion unlike compression takes place smoothly through infinite expansion waves hence called as expansion fan. This expansion fan is comprised of infinite number of expansion waves or Mach waves where every wave is responsible for infinitesimal amount of deflection. Typical expansion fan in the supersonic flow is shown in **Fig.** where supersonic flow turns outward by an angle θ .



Expansion of supersonic flow

For better understanding of expansion of supersonic flow, consider the that p_1 , T_1 and M_1 be the properties of flow before expansion or upstream of the expansion fan and p_2 , T_2 and M_2 be the properties of the flow after expansion or downstream of the expansion fan due to outward deflection by an angle θ . For the know upstream flow properties and deflection angle it should be possible for us to calculate the downstream flow properties. Since the expansion is the continuous and smooth process carried out via infinite Mach waves, let's consider one such wave across upstream of which velocity is V and Mach number is M . Angle made by this Mach wave with the upstream velocity vector is μ . Let's

consider dV be the change in velocity brought by the Mach wave by turning through an angle $d\theta$. Hence $V+dV$ is downstream velocity and $M+dM$ is the downstream Mach number. If we construct the velocity triangle as shown in **Fig.** then we can use the sin law for triangle as,



Velocity triangle across a typical Mach wave during supersonic expansion

$$\frac{V + dV}{V} = \frac{\sin\left(\frac{\pi}{2} + \mu\right)}{\sin\left(\frac{\pi}{2} - \mu - d\theta\right)}$$

But we know that

$$\sin\left(\frac{\pi}{2} + \mu\right) = \sin\left(\frac{\pi}{2} - \mu\right) = \cos \mu$$

and

$$\sin\left(\frac{\pi}{2} - \mu - d\theta\right) = \cos(\mu + d\theta) = \cos \mu \cos d\theta - \sin \mu \sin d\theta$$

Hence we can re-write **Eq. (25.1)** as,

$$1 + \frac{dV}{V} = \frac{\cos \mu}{\cos \mu \cos d\theta - \sin \mu \sin d\theta}$$

We can approximate as

$\sin d\theta \approx d\theta$ and $\cos d\theta \approx 1$, Therefore Eq. (25.2) can be simplified as,

$$1 + \frac{dV}{V} = \frac{\cos \mu}{\cos \mu - d\theta \sin \mu} = \frac{1}{1 - d\theta \tan \mu}$$

since $d\theta \tan \mu < 1$, let's recall the expansion for $x < 1$,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

Neglecting higher order term, we can express Eq. (25.3) as,

$$1 + \frac{dV}{V} = 1 + d\theta \tan \mu + \dots$$

$$d\theta = \frac{dV/V}{\tan \mu}$$

But we know that

$$\mu = \sin^{-1} \frac{1}{M} \quad \text{and hence} \quad \tan \mu = \frac{1}{\sqrt{M^2 - 1}}$$

Hence above equation becomes,

$$d\theta = \sqrt{M^2 - 1} \frac{dV}{V}$$

Prandtl Mayer Function

We can see that since for positive value of $d\theta$, we get positive dV which leads to expansion. This formula is also valid for small angles for compression where we get negative dV . If we integrate this formula for the total expansion angle then we can get the downstream Mach number.

$$\int_{\theta_1}^{\theta_2} d\theta = \int_{M_1}^{M_2} \sqrt{M^2 - 1} \frac{dV}{V}$$

Before integrating we can express the integrant in Mach number,

$$V = Ma$$

$$\ln V = \ln M + \ln a$$

$$\frac{dV}{V} = \frac{dM}{M} + \frac{da}{a}$$

We can express here the second term on right hand side in terms of Mach number using the isentropic relations as,

$$\left(\frac{a_o}{a}\right)^2 = \frac{T_o}{T} = 1 + \frac{\gamma-1}{2} M^2$$

$$a = a_o \left(1 + \frac{\gamma-1}{2} M^2\right)^{-1/2}$$

$$\frac{da}{a} = -\left(\frac{\gamma-1}{2}\right) M \left(1 + \frac{\gamma-1}{2} M^2\right)^{-1} dM$$

Using Eq. (25.5) and (25.6) we can re-write Eq. (25.4) as,

$$\int_{\theta_1}^{\theta_2} d\theta = \theta_2 - \theta_1 = \int_{M_1}^{M_2} \frac{\sqrt{M^2-1}}{1 + \frac{\gamma-1}{2} M^2} \frac{dM}{M}$$

Integration of right hand side is as,

$$v(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1} \sqrt{\frac{\gamma-1}{\gamma+1} (M^2-1)} - \tan^{-1} \sqrt{M^2-1}$$

Here, v is called as the Prandtl-Meyer function.

$$\theta = v(M_2) - v(M_1)$$

Therefore upstream Mach number (M_1) we can calculate the upstream Prandtl-Meyer function. Hence for known flow deflection angle and upstream Mach number we can get the downstream Prandtl-Meyer function and hence the downstream Mach number.

Process of expansion of supersonic flow is an isentropic process. However, while

passing through the expansion fan, pressure, temperature and density of the flow decreases while Mach number and velocity increases for the supersonic flow. Moreover, all the total properties remain constant. We can calculate the total pressure, temperature and density upstream of the expansion using isentropic relations for the known flow Mach number. From the calculated downstream Mach number, we can calculate all the static flow properties from known stagnation or total properties.